Supervised Bayesian factor models for molecular programs and survival in pancreatic cancer

Technical report on the evaluated methods, current evidence, and remaining work

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Status: initial draft awaiting independent review. This report describes methods and evidence available through October 12, 2026. It has not yet been reviewed independently, and it is not ready to share beyond the immediate project team. Every quantitative statement is traced to a saved output in `paper/writeup/source_inventory.csv`; claims that could not be traced are omitted or listed as evidence gaps. The planned manuscript authors are Walther, Liu, and Rashid. DeSurv (Young et al. 2026) is cited throughout as the source of the gene-screening rule and the published factor gene lists.

Summary of current findings

What the models are. Both models factor molecular data into nonnegative patient activities L and feature loadings F, and link survival to the observable projection of each patient's data onto the loadings, $\eta = (YF)\beta$. The single-modality model fits one feature block. The multimodal model

shares L across matched expression and methylation blocks, adds per-feature intercepts and signed loadings, and propagates loading uncertainty into its coefficient updates.

What the evaluated single-modality configuration does. In the configuration behind all single-modality results ($\alpha_F = 0$), survival informs the coefficients β but not the loadings F or the activities L. The programs are therefore the programs of an empirical-Bayes factorization of the training data; survival selects and weights them. This is not the intended final method (Section 5.4).

Single-modality performance. Trained on TCGA-PAAD RNA-seq and CPTAC proteomics ($n = 273$), the frozen model has a mean concordance of 0.627 across five held-out cohorts ($n = 616$). Against an independently tuned two-step comparator (40 unsupervised factors, LASSO Cox), the C-index difference computed on the 616 concatenated patients is 0.026 (95% bootstrap interval 0.000 to 0.050). Against a two-step comparator given the same rank of 7 the difference is 0.011 (-0.013 to 0.033).

Single-modality biology. Starting from rank 7, four programs are retained and two are classified as survival-active. The adverse program (7) aligns with the basal-like axis and the protective program (3) with the classical axis; the two genomics-only programs read as desmoplastic stroma (5) and immune infiltrate with a vascular component (6). These labels rest on enrichment, subtype concordance and gene-list overlap analyses that share inputs and were not prespecified.

Multimodal model. In simulation, the framework with intercepts, signed loadings, empirical-Bayes priors and evidence-based pruning returns close to the true rank of 3 from every starting rank at moderate noise, and converges in all 162 of its fits. It does not recover a planted low-variance prognostic program at moderate noise. On real data, nothing is pruned at any starting rank, most fits reach the 500-sweep cap, and external concordance on 50 primary-PDAC ICGC donors varies from 0.50 to 0.69 across starting ranks. At starting rank 7 the joint fit has $C = 0.685$ and the expression-only fit of the same framework $C = 0.676$; both are unconverged. Methylation adds a methylation-dominant program and loads on an immune program; it does not demonstrably improve prediction.

Main open problems. Survival-guided discovery of programs, a rank-selection rule that accounts for both survival and reconstruction on real data, full convergence of real multimodal fits, feature-count sensitivity, matched comparisons with DeSurv and unsupervised factorization, and a methylation-specific biological interpretation (Section 5.2).

1 Methods

1.1 Notation

Patients are indexed by $i = 1, \ldots, n$, molecular features by j, factors (programs) by $k = 1, \ldots, K$, and data blocks (modalities) by m. Patient i has observed time t_i and event indicator δ_i (1 for death, 0 for censoring). The single-modality data are an $n \times p$ matrix Y; the multimodal data are matched blocks Y_m of dimension $n \times p_m$, here expression ($m = \text{expr}$) and DNA methylation ($m = \text{meth}$). Methylation measurements are written Y_{meth}; the symbol β is reserved for survival coefficients.

Three patient-level quantities must be kept apart: the **latent activities** $L \in \mathbb{R}^{n \times K}_{\geq 0}$, posterior scores in the reconstruction $Y \approx LF^\top$, defined only for training patients; the **projection scores**, the observable product of a patient's data with the posterior-mean loadings, which enter the Cox

predictor and can be computed for new patients ($\tilde{Z}$ in Equation 2, Z in Equation 5); and the **signature scores** used only in biological characterization, the mean of within-cohort z-scored expression over a selected gene set.

Table 1: Model and notation summary. Settings are those of the evaluated fits, not package defaults. $\bar{F}$ denotes posterior means, $D = \mathrm{diag}(\|\bar{F}_{\cdot k}\|_2)$, and $\tilde{Y}_m$ is Y_m centered at the current intercept. Sources: S04, S08, S29, S37.

	Single-modality model (evaluated configuration D4)	Multimodal model (evaluated framework)
Data	One block Y $(n \times p)$; training on TCGA-PAAD RNA-seq and CPTAC proteomics, per-platform standardized	Matched blocks Y_{expr}, Y_{meth}; TCGA-PAAD, log2 expression and methylation β-values
Reconstruction	$Y = LF^\top + E$	$Y_m = \mathbf{1}\mu_m^\top + LF_m^\top + E_m$, shared L
Activity prior L_{ik}	Point-exponential, fitted per factor by empirical Bayes	Point-exponential, fitted per factor by marginal likelihood (`ebnm`)
Loading prior	Point-exponential ($F \geq 0$)	Point-Laplace per modality and factor (signed F_m)
Residual precision	Feature-specific τ_j, maximum likelihood	Feature-specific τ_{mj}, maximum likelihood
Cox predictor	$\eta = \tilde{Z}\tilde{\beta}$, $\tilde{Z} = Y\bar{F}D^{-1}$	$\eta = Z\beta$, $Z = \sum_m \tilde{Y}_m F_m$
Coefficient prior	Normal, variance estimated per factor	Normal, variance estimated per factor
Survival in the loading update	None ($\alpha_F = 0$)	Yes, equal weight, with feature coupling
Uncertainty in the coefficient update	Projection treated as fixed (plug-in)	Projection mean and variance from $q(F_m)$
Scale handling	Loadings normalized to unit length inside $\tilde{Z}$	Each factor rescaled so that $\mathrm{sd}(Z_k) = 1$ after every update
Stopping rule	Relative change $< 10^{-5}$ in a weighted monitoring objective; cap 300	Relative change $< 10^{-4}$ in reconstruction and η; cap 500 sweeps
Rank handling	Start at $K = 7$; classify factors after fitting	Start at K_{init}; remove factors by an evidence check, then refit
New-patient prediction	$\eta_{\mathrm{new}} = y_{\mathrm{new}}^\top \bar{F} D^{-1}\tilde{\beta}$, frozen orientation	$\eta_{\mathrm{new}} = \sum_m y_{m,\mathrm{new}}^\top \bar{F}_m \beta$, frozen orientation

1.2 Single-modality model

Reconstruction and priors. The training matrix is modeled as

$$y_{ij} = \sum_{k=1}^{K} L_{ik}F_{jk} + e_{ij}, \qquad e_{ij} \sim N(0, \tau_j^{-1}), \tag{1}$$

with independent point-exponential priors $L_{ik} \sim g_{L,k}$ and $F_{jk} \sim g_{F,k}$ (a point mass at zero mixed with an exponential distribution), whose parameters are estimated by empirical Bayes within each factor using the `ebnm` package. Each feature has its own residual precision τ_j, estimated by maximum likelihood.

Survival predictor. Survival follows a Cox model whose linear predictor is the normalized projection of the patient's data onto the posterior-mean loadings:

$$\eta_i = \sum_{k=1}^{K} \tilde{Z}_{ik}\tilde{\beta}_k, \qquad \tilde{Z} = Y\bar{F}D^{-1}, \quad D = \mathrm{diag}(\|\bar{F}_{\cdot 1}\|_2, \ldots, \|\bar{F}_{\cdot K}\|_2). \tag{2}$$

Each column of $\bar{F}D^{-1}$ has unit Euclidean length, so $\tilde{Z}_{ik}$ is the coordinate of y_i along program k's loading direction. The normalization keeps $\tilde{Z}$ on a scale that does not grow with p; without it, coefficient shrinkage drove every $\tilde{\beta}_k$ to zero (S04). The coefficients $\tilde{\beta}_k$ have independent normal priors with zero mean and a variance estimated per factor, so they are interpreted per unit of normalized projection, not per standard deviation. Because $\tilde{Z}$ depends only on Y and $\bar{F}$, the same formula scores training and new patients.

Cox working approximation. The Cox partial likelihood is replaced at each sweep by a second-order expansion around the current predictor $\hat{\eta}$ with a diagonal Breslow curvature,

$$\log \mathrm{PL}(\eta) \approx \mathrm{const} - \tfrac{1}{2}\sum_i w_i(z_i - \eta_i)^2, \qquad w_i = \hat{\theta}_i \hat{H}_0(t_i), \quad z_i = \hat{\eta}_i + (\delta_i - w_i)/w_i, \tag{3}$$

where $\hat{\theta}_i = \exp(\hat{\eta}_i)$ and $\hat{H}_0$ is the Breslow cumulative baseline hazard. Off-diagonal curvature is dropped. Appendix A derives each coordinate update from this surrogate.

Where survival enters. Two weights appear in the implementation. The survival weight $\alpha \in [0, 1]$ multiplies the survival term in the coefficient update; the evaluated value is $\alpha = 0.5$, which leaves the pseudo-observation B/A unchanged and halves its precision. The loading weight α_F mixes a reconstruction contribution, with weight $1 - \alpha_F$, and a survival contribution, with weight α_F, in the update of F. The evaluated configuration uses $\alpha_F = 0$. The activities L do not appear in η, so their update is a reconstruction update in every configuration. Consequently, in the evaluated fits survival determines $\tilde{\beta}$ but not L or F; it affects which programs are scored as prognostic, not what the programs are. The coefficient update also treats $\tilde{Z}$ as fixed: it does not propagate posterior uncertainty in F (Table 12).

1.3 Single-modality fitting algorithm and prediction

Algorithm 1. Evaluated single-modality fit (`fit_cox_on_yf()`, configuration D4)

Input: standardized training matrix Y ($n = 273$, $p = 2{,}064$), (t_i, δ_i), starting rank $K = 7$, $\alpha = 0.5$, $\alpha_F = 0$, normal coefficient prior.

1. **Initialize** $\bar{L}, \bar{F}$ from the absolute values of the leading K singular vectors of Y scaled by $\sqrt{d_k}$; set $\tilde{\beta} = 0$ and $\tau_j = 1/\widehat{\mathrm{var}}(y_{\cdot j})$.
2. **For each outer iteration** (at most 300):
   a. Form D and $\tilde{Z} = Y\bar{F}D^{-1}$; compute $\eta = \tilde{Z}\tilde{\beta}$ and the working quantities (w, z) of Equation 3. These are held fixed for the rest of the iteration.
   b. **For** $k = 1, \ldots, K$**:**
      i. update $q(\tilde{\beta}_k)$ from $A = \alpha\sum_i w_i \tilde{Z}_{ik}^2$ and $B = \alpha\sum_i w_i \tilde{Z}_{ik}(z_i - \eta_{i,-k})$ by empirical-Bayes normal means with pseudo-observation $x = B/A$ and standard error $A^{-1/2}$;
      ii. update $q(L_{\cdot k})$ from the reconstruction residual excluding factor k (no survival term);
      iii. update $q(F_{\cdot k})$ from the reconstruction residual only ($\alpha_F = 0$).
   c. Update $\tau_j = n/\sum_i E(y_{ij} - L_i^\top F_j)^2$.
   d. Compute the monitoring objective $J = (1-\alpha)G + \alpha pS - \mathrm{KL}$ (Appendix A.8). Stop after iteration 5 if $|\Delta J|/|J| < 10^{-5}$.

3. **Orientation.** Compare the training concordance of η with 0.5 and, if needed, negate $\bar{\tilde{\beta}}$ once. The saved D4 fit predates the September 4 correction of this check, so its η is a good-prognosis score; it is scored as risk $= -\eta$ for every held-out cohort (Section 1.3.1).
4. **Classify** factors (Section 1.4).

Output: $\bar{L}$, $\bar{F}$, D, $\bar{\tilde{\beta}}$, τ. **Prediction:** for a new patient with standardized data y_{new} restricted to the training genes that are measured, $\text{risk}_{\text{new}} = -\, y_{\text{new}}^{\top} \bar{F} D^{-1} \bar{\tilde{\beta}}$, with all quantities fixed from training.

The saved D4 fit met the stopping rule at iteration 20 (S04, S58). The monitoring objective weights the reconstruction and survival terms by $1-\alpha$ and αp; it is used to decide when to stop and is not the evidence lower bound of a single generative model. The coefficient posterior variances stored with this historical fit equal zero to machine precision, consistent with the pre-September sign-correction code, which overwrote $E(\tilde{\beta}^2)$ with $E(\tilde{\beta})^2$ when it negated the coefficients (S01, 2026-09-04 entry). The report therefore gives no coefficient uncertainty for this fit.

1.3.1 Risk orientation

A concordance index requires a direction: which end of the score indicates higher risk. That direction must be fixed from training data alone; choosing it in each validation cohort from that cohort's outcomes would bias the index upward. For the saved D4 fit, $\eta = \tilde{Z}\tilde{\beta}$ is concordant with longer survival, and `make_km_figures.R` and `run_external_ci_analysis.R` use the single global convention risk $= -\eta$ for all held-out cohorts (S01, S49). On the risk scale, the coefficient of Program 7 is $+0.040$ and that of Program 3 is -0.012. This convention is specific to the historical saved fit and is not a rule for current fits, whose orientation step was corrected on September 4.

1.4 Programs, starting rank, and classification

Three counts must be kept apart. The **starting rank** K_{init} is the number of factors at initialization. A factor is **survival-active** if $|\bar{\tilde{\beta}}_k| > 0.001$. A factor that is not survival-active is **genomics-only** if its proportion of variance explained, $\text{PVE}_k = \|\bar{L}_{\cdot k}\|^2 \|\bar{F}_{\cdot k}\|^2 / \|Y\|_F^2$, exceeds 1%, and **dead** otherwise. The **retained molecular programs** are the survival-active and genomics-only factors together. In the single-modality model this is a classification of one fitted model: no factor is removed and nothing is refitted. The two thresholds are operational. The coefficient threshold was set from the observed scale of $\tilde{\beta}$ in this model (S37), and neither threshold is a posterior probability.

The starting rank of 7 has two sources. It was first chosen by five-fold cross-validated training concordance with a one-standard-error rule: the best mean was at $K = 8$ (0.651, SE 0.026), and $K = 7$ was the smallest rank within one standard error (0.633; S55). It was retained after a sweep over $K_{\text{init}} = 2, \ldots, 20$ in which the criteria disagreed (Section 3.3). External concordance in the five held-out cohorts reaches a plateau at $K_{\text{init}} = 7$, and that plateau was part of the evidence for keeping 7. The recommendation is therefore not purely training-selected.

1.5 Multiple cohorts in the single-block model

The established model fits one feature block pooled from two training cohorts on different platforms: TCGA-PAAD RNA-seq ($n = 144$) and CPTAC mass-spectrometry proteomics ($n = 129$). CPTAC

is a proteomics cohort, not a second RNA-seq cohort (S03). Genes are screened separately in each cohort by the DeSurv combined mean-and-variance rank (Young et al. 2026), keeping the top 3,000 per cohort; the intersection has 2,064 genes. Each cohort is then standardized gene by gene to mean 0 and standard deviation 1 (per-platform standardization). Held-out cohorts receive the same per-platform standardization and are scored on the training genes they measure.

Three optional cohort terms are distinct model components: `cohort_id` adds cohort indicator columns to L (a genomic offset), `strata_id` forms Cox risk sets within cohort (a cohort-specific baseline hazard), and `beta_cohort_id` gives each cohort its own partially pooled coefficients. The recommended configuration uses none of them; the decision record reports the last two as performance-neutral externally (S01).

1.6 Multimodal model

Reconstruction. For matched blocks $m \in \{\text{expr}, \text{meth}\}$,

$$Y_m = \mathbf{1}\mu_m^\top + LF_m^\top + E_m, \qquad (E_m)_{ij} \sim N(0, \tau_{mj}^{-1}), \tag{4}$$

with a feature intercept μ_m re-estimated at each outer sweep, shared nonnegative activities L (point-exponential prior per factor), modality-specific signed loadings F_m with a point-Laplace prior g_{mk} for each modality and factor, and feature-specific precisions τ_{mj}. The intercept is what makes signed loadings appropriate: F_m describes deviations from each feature's baseline, and L remains interpretable as program activity. Prior parameters for L and F_m are estimated by marginal likelihood after each sweep, so a whole column can shrink to the point mass at zero.

Survival predictor. With $\tilde{Y}_m = Y_m - \mathbf{1}\mu_m^\top$, the joint projection and predictor are

$$Z = \sum_m \tilde{Y}_m F_m, \qquad \eta_i = \sum_k Z_{ik}\beta_k, \qquad \beta_k \sim N(0, \sigma_k^2). \tag{5}$$

The projection is not normalized; instead, after every factor update, $(L_{\cdot k}, F_{mk}, \beta_k)$ is replaced by $(c_k L_{\cdot k}, F_{mk}/c_k, c_k\beta_k)$ with $c_k = \text{sd}(Z_{\cdot k})$, which leaves both $LF_m^\top$ and η unchanged and fixes $\text{sd}(Z_{\cdot k}) = 1$ in training. This canonicalization removes the scale direction along which survival and reconstruction terms can otherwise drift (S01, 2026-10-02 entry). Posterior second moments and prior rates are transformed under the same change of variables. A constant shift of η leaves the partial likelihood unchanged, so the intercept does not affect the survival term.

Updates. The loading update adds reconstruction and Cox working terms with equal weight and couples features through the projection, so features are updated one at a time with the projection moments refreshed after each. The coefficient update uses $E(Z_{ik}^2) = E(Z_{ik})^2 + \text{Var}(Z_{ik})$, so loading uncertainty shrinks the coefficient. Appendix B gives the derivations.

Evidence and factor removal. The objective used for factor removal is

$$\mathcal{E} = \sum_{m,j} \left[\tfrac{n}{2} \log \tfrac{\hat{\tau}_{mj}}{2\pi} - \tfrac{\hat{\tau}_{mj}}{2} E\|\tilde{y}_{mj} - Lf_{mj}\|^2 \right] + \log \text{PL}(E(\eta)) - \text{KL}(q \,\|\, g), \tag{6}$$

with $\hat{\tau}$ at its maximum-likelihood value. The survival term is the partial log-likelihood at the posterior mean predictor (a plug-in), not its expectation under q. After the coordinate loop, the change in $\mathcal{E}$ from deleting each factor is computed with the other factors fixed and the intercept and

precisions re-estimated; if a deletion does not lower $\mathcal{E}$, the factor with the largest gain is removed and the model refitted, repeatedly. The check runs whether or not the loop converged (S08).

Classification. A factor is reconstruction-active if its share of total sum of squares, $\sum_m \|\bar{L}_{\cdot k}\bar{F}_{mk}^\top\|_F^2 / \sum_m \|\tilde{Y}_m\|_F^2$, is at least 1%, and survival-active if $|E\beta_k|/\mathrm{sd}(\beta_k) \geq 1.96$. These are reported separately from removal.

1.7 Multimodal fitting algorithm

Algorithm 2. Evaluated multimodal fit (`fit_multimodal_yfb()`, pruned framework)

Input: raw training blocks Y_m (log2 expression, methylation β-values), (t_i, δ_i), starting rank K_{init}. Controls: intercept on; point-Laplace loading priors; marginal-likelihood prior updates; feature-specific maximum-likelihood τ; plug-in survival term; pruning on; at most 500 outer sweeps of 5 inner sweeps; tolerance 10^{-4}.

1. **Initialize** μ_m at column means; $\bar{L}$ and $\bar{F}_m$ from absolute singular vectors of the concatenated centered blocks; τ_{mj} from the pooled variance of each block; β from a Cox fit on standardized initial projections, transformed back to the raw projection scale.
2. **For each outer sweep** (at most 500):
   a. Re-estimate μ_m and re-center $\tilde{Y}_m$. Compute $E(Z)$, $\mathrm{Var}(Z)$, $E(\eta)$, and the Cox working quantities w_i and $h_i = u_i + w_i E(\eta_i)$; hold (w, h) fixed for this outer sweep.
   b. **Repeat 5 inner sweeps; for** $k = 1, \ldots, K$**:**
      i. update $q(L_{\cdot k})$ from both blocks' reconstruction residuals; refit its point-exponential prior;
      ii. for each modality, update $q(f_{mjk})$ feature by feature with reconstruction and survival terms, refreshing $E(Z_{\cdot k})$ and $\mathrm{Var}(Z_{\cdot k})$ after each feature; refit the point-Laplace prior g_{mk};
      iii. update $q(\beta_k)$ with $A = \sum_i w_i E(Z_{ik}^2)$, $B = \sum_i E(Z_{ik}) h_{i,-k}$, and prior variance $\sigma_k^2 = \max(0, B^2/A^2 - 1/A)$;
      iv. canonicalize factor k so that $\mathrm{sd}(Z_{\cdot k}) = 1$.
   c. Update $\tau_{mj} = n / E\|\tilde{y}_{mj} - L f_{mj}\|^2$.
   d. Stop if the maximum relative change in each fitted reconstruction $\bar{L}\bar{F}_m^\top$ and in $E(\eta)$ is below 10^{-4}.
3. **Removal check.** Compute the change in Equation 6 from deleting each factor; if the largest change is ≥ 0, delete that factor, refit from the remaining factors (returning to step 2 with a warm start), and repeat.

Output: posterior moments, priors, μ_m, τ, classification. **Prediction:** $\mathrm{risk}_{\text{new}} = \sum_m y_{m,\text{new}}^\top \bar{F}_m \bar{\beta}$, with orientation fixed from training (`flip = FALSE`). Centering new data at the training intercept would change every score by the same constant and leave the C-index unchanged.

1.8 Approximations and what the objectives do not guarantee

Each coordinate update is exact for the quadratic Cox surrogate at fixed working quantities, but the surrogate changes between sweeps, its curvature is diagonal, the single-modality monitor reweights its components, and the multimodal removal objective evaluates survival at $E(\eta)$. Neither algorithm is therefore exact coordinate ascent on one evidence lower bound, and neither is guaranteed to

increase monotonically. Appendices A and B list each approximation beside the update it affects.

The evaluated single-modality configuration with $\alpha_F = 0$ is a stage in the development of the method, not its intended final form. Planned work on survival supervision of the loadings, coefficient priors, and rank selection is described in one place, Section 5.4.

2 Evaluation framework

2.1 Simulation designs

Single-modality, multiple cohorts (S17, S18, S59). Each dataset has two cohorts of 150 patients and 1,000 genes. True loading columns are the absolute values of unsupervised factorization programs estimated from the real training data, scaled to amplitude 12; feature noise precisions are drawn from Gamma(2, 2). Survival times follow a Weibull proportional-hazards model with predictor $\sum_k L_{ik}\beta_k$ over the shared programs (coefficients recycled from $1.5, -1.2, 0.8, -0.5$), and censoring is calibrated to 30%. Study-specific programs have zero loadings outside their cohort and no survival effect. Three scenarios vary the structure: all shared (six shared programs), hybrid (two shared and two specific per cohort), and nothing shared (three specific per cohort, so no program is prognostic). Each dataset is split 75/25 into training and test patients, stratified by event status. Two arms are fitted at starting ranks 2, 3, 4, 6, 12, and 20 with 15 seeds, giving 540 fits: the single-modality model with the evaluated settings, and a two-step arm consisting of a nonnegative empirical-Bayes factorization (`flashier`, at most K_{init} factors, backfitted) followed by an unpenalized Cox model on least-squares projection scores. Note that the data-generating predictor uses the latent activities L, whereas the model predicts from projections YF.

Multimodal (S25, S26, S59). Each dataset has 120 training and 120 validation patients, 50 expression and 50 methylation features, and three true programs. Survival follows a Weibull model with predictor $\sum_m Y_m F_m \beta$, matching the model's predictor form. Three scenarios set β: two prognostic programs $(0.70, 0.35, 0)$, adverse and protective $(0.70, -0.70, 0)$, and a low-variance prognostic program whose activities are scaled by 0.15 ($\beta = 0, 0, 1.10$). Noise is set by a multiplier on every feature's noise standard deviation: 1 reproduces the September design and 0.25 is lower noise ("moderate"). Each feature is shifted by four noise standard deviations so the data are nonnegative and the original model can be fitted. Starting ranks are 3–10 and 15, with three seeds per setting. Three variants are fitted to every dataset: the original September model (no intercept, nonnegative loadings, moment prior updates, no pruning), the evaluated pruned framework (Algorithm 2), and the same framework with the variance-corrected survival term. The full sweep contains 486 fits, 162 per variant.

2.2 Data

Table 2: Cohorts and preprocessing. Events in the two single-modality training cohorts are reported pooled (139 deaths among 273 patients). The 50 matched ICGC donors overlap the PACA-AU RNA-seq external cohort and are not an additional independent replication; whether the PACA-AU array and RNA-seq cohorts share donors was not checked in the reviewed sources. Sources: S03, S09, S11, S30, S01 (2026-10-01 entry), and the orientation check recorded under S56.

Analysis	Cohort	Platform	Patients	Events	Role and preprocessing
Single-modality	TCGA-PAAD	RNA-seq	144	139 (pooled)	Training; log2(count+1); screened and standardized within cohort
	CPTAC	Mass-spectrometry proteomics	129		Training; provided normalized scale; screened and standardized within cohort
	Dijk	RNA-seq	90	81	External validation
	Moffitt	Microarray	123	83	External validation
	PACA-AU array	Microarray	63	38	External validation
	PACA-AU RNA-seq	RNA-seq	52	31	External validation
	Puleo	Microarray	288	181	External validation
	Five external cohorts		616	414	Each standardized within cohort; scored on 1,989–2,050 of the 2,064 training genes
Multimodal	TCGA-PAAD	RNA-seq + 450K methylation	144	75	Training; 3,000 genes (DeSurv rank) and 10,000 highest-variance CpGs, screened on TCGA only
	ICGC PACA-AU, primary PDAC	RNA-seq + methylation	50	30	Main validation: all samples primary tumors with PDAC histology
	ICGC PACA-AU, all donors	RNA-seq + methylation	67	40	Sensitivity: adds IPMN with invasion, adenosquamous, acinar and signet-ring tumors

The matched multimodal cohorts are built from the TCGA and ICGC files with expression and methylation for the same patients (S30). Methylation is restricted to the 305,352 CpGs present in both cohorts' sex-chromosome-filtered probe sets; missing values (0.07% in TCGA) are imputed within cohort by k-nearest neighbors. The evaluated multimodal fits use raw inputs: log2 expression and methylation β-values. Screening uses training data only.

2.3 Comparators and prediction rules

Principal single-modality comparator. An unsupervised empirical-Bayes factorization (`flashier`, signed point-normal priors, gene-specific variances, greedy fitting with backfitting, at most 40 factors) of the same $273 \times 2{,}064$ training matrix, followed by a LASSO Cox model on projection scores $Y\bar{F}D^{-1}$ with the penalty chosen by cross-validation (`lambda.min`). The factor count was not informed by the supervised model; 15 of the 40 factors received nonzero coefficients (S10). Held-out patients are scored with the frozen loadings and coefficients.

Supporting single-modality comparators. A rank-matched two-step model (7 factors, unpenalized Cox; S12) and the earlier comparator (20 factors, unpenalized Cox; S13). Matching the rank gives the two-step model the supervised model's answer about complexity, and the un-

penalized 20-factor model is not regularized; neither is the principal comparison. The output directory `desurv_comparison` contains our own models fitted with DeSurv-aligned and alternative preprocessing (configurations D1–D5); it is not a fitted DeSurv benchmark (S48).

Multimodal comparators. On the matched cohorts, the same framework fitted to expression alone (configuration `pruned_expression_only`) isolates the contribution of methylation. Two further arms are reported for completeness: an older isolated single-modality fitter on raw log2 expression (`expression_only_yfb`, which is not the expression-only fit of the evaluated framework) and two-step factorization followed by Cox on stacked or expression-only data (S28). The established single-modality model was trained on different data and preprocessing and is discussed separately (Section 3.5).

2.4 Performance metrics and uncertainty

External performance is Harrell's concordance index with a risk orientation fixed from training. All per-cohort single-modality values in Table 3 were re-computed during drafting from the saved risk scores with frozen orientation and agree with the saved values to four decimals (S56); some of the saved runners had computed the outcome-oriented $\max(C, 1-C)$, which coincides with the frozen value in every cohort here. Confidence intervals are percentile bootstrap intervals over patients ($B = 2{,}000$ for single-modality, 1,000 for multimodal; seed 1).

Three summaries of the single-modality comparison are not interchangeable: the **mean external C** averages five cohort C-indices; the **concatenated difference** (principal and rank-matched comparisons) computes one C-index per model over all 616 patients, so comparable pairs cross cohorts, with an unstratified bootstrap; and the **inverse-variance pooled difference** (older comparator) combines per-cohort paired differences. None is an equivalence or non-inferiority test.

Simulation summaries are means over seeds. In the single-modality simulation, test-set concordance is the outcome-oriented $\max(C, 1-C)$ (S17), which is biased upward when there is no signal: in the nothing-shared scenario both arms score about 0.54. In the multimodal simulation, validation concordance uses the frozen orientation. Program recovery is the absolute correlation between a true loading column and its best-matching estimated column; a program counts as recovered at $|r| \geq 0.7$.

2.5 Biological characterization methods

The biological labels in Section 4 come from existing analyses of the established single-modality fit and of the starting-rank-7 multimodal fit. Table 14 in Appendix C lists, for each analysis, the object analysed, the population, the statistic and settings, the multiplicity scope, and what the output can support. No analysis in this table was rerun for this report.

Several of these analyses share inputs and are not independent validations. The custom fgsea collection and the gene-list overlap both use DeSurv's lists, and the selected gene universe was itself screened with DeSurv's rule. The subtype concordance uses training activities $E(L)$ in TCGA, which also trained the model. The original external signature check (C2) scored each program's lowest-padj set; for Program 6 that set was the DeSurv classical-tumor list, the same list used as Program 3's signature, so C2 measured the classical axis rather than Program 6. The corrected control (C4) excludes the entire PDAC_custom collection, including the Moffitt and Bailey lists, and

was designed after C2's result; it is a post hoc correction. Multiple-testing control is within program and collection only. The pathway report itself states that these methods were chosen for availability rather than from a review of what the question requires and that the choice warrants assessment before manuscript use (S21). Two further cautions apply to the signature helper: its hazard ratio is per unit of the mean z-scored signature, not per standard deviation, and its concordance output is $\max(C, 1 - C)$, which is outcome-oriented and cannot substitute for frozen-risk validation. Those concordance values are not used in this report.

3 Results

3.1 Single-modality simulation

Table 15 and Figure 5 in Appendix C summarize the 540 fits. At or above the true rank (6), the model recovers every shared program at $|r| \geq 0.7$ and nearly every study-specific program, and over-specifying the rank to 12 or 20 changes little. The two-step arm recovers fewer programs and stops changing beyond rank 6 because its greedy factorization stops adding factors. The model's test concordance exceeds the two-step arm's by about 0.10 in the hybrid scenario from rank 3 upward and by about 0.02 in the all-shared scenario from rank 6; with nothing prognostic both are near the outcome-oriented null of 0.54. With $\alpha_F = 0$ the model's L and F updates ignore survival, so these differences come from the fitting procedure (singular-vector initialization and coordinate updates versus greedy `flashier`), not from survival supervision. Possible merges, an estimate correlated above 0.5 with two true programs, occurred in a single-seed diagnostic only at starting ranks 2–4 (S19). Among programs without a survival effect, 22%–35% were classified survival-active in the hybrid scenario and 6%–10% with nothing shared.

A separate ten-seed simulation tested cohort-specific coefficients for a program that is prognostic in only one cohort (S20). The coefficient in the owning cohort exceeded the other cohort's in 15 of 20 cohort-by-seed cases at starting rank 6 and 17 of 20 at rank 12 (medians 0.060 versus 0.008, and 0.054 versus 0.008); one seed diverged, so medians are the reported summary.

3.2 Single-modality external validation

Table 3 gives per-cohort concordance. The frozen model's mean external C is 0.627 (0.549 in Moffitt to 0.657 in PACA-AU RNA-seq) and the principal comparator's 0.601. Over the 616 concatenated patients the C-indices are 0.629 and 0.603, a difference of 0.026 with 95% interval 0.000 to 0.050; the lower limit is 0.0002, so the advantage is marginal (S11, S56, S60). The rank-matched comparator, which is better in both PACA-AU cohorts, gives a difference of 0.011 (−0.013 to 0.033; S12). The older unpenalized comparator gives 0.042 (0.013 to 0.071), with a per-cohort interval excluding zero only in Puleo (S13).

Table 3: External concordance of the frozen single-modality model (risk = $-\eta$) and two-step comparators. All C-indices use training-fixed orientation (S56). [a]Difference of C-indices computed on the 616 concatenated patients, bootstrap over patients. [b]Inverse-variance pooled per-cohort paired differences. The 40-factor LASSO model is the principal comparator; the rank-matched and 20-factor unpenalized models are supporting. Sources: S09, S10, S11, S12, S13, S56, S57, S60.

Cohort	n	Events	Model C (95% CI)	Two-step, 40 factors, LASSO	Two-step, 7 factors	Two-step, 20 factors
Dijk	90	81	0.634 (0.555, 0.708)	0.600	0.577	0.592
Moffitt	123	83	0.549 (0.467, 0.625)	0.560	0.541	0.541
PACA-AU array	63	38	0.648 (0.553, 0.747)	0.623	0.665	0.597
PACA-AU RNA-seq	52	31	0.657 (0.544, 0.772)	0.603	0.690	0.573
Puleo	288	181	0.645 (0.599, 0.688)	0.618	0.643	0.602
Mean of five	616	414	0.627	0.601	0.623	0.581
Difference, model minus comparator				0.026 (0.000, 0.050)[a]	0.011 (−0.013, 0.033)[a]	0.042 (0.013, 0.071)[b]

Within each cohort, patients were split into tertiles of the frozen risk score (Figure 1). The low-risk tertile has the longest survival in every cohort; the log-rank test has $p < 0.05$ in Dijk ($p = 0.00046$), PACA-AU RNA-seq (0.021), and Puleo (< 0.0001), and not in PACA-AU array (0.075) or Moffitt (0.17) (S36). Tertiles are within-cohort descriptive groups, not fixed clinical cutoffs.

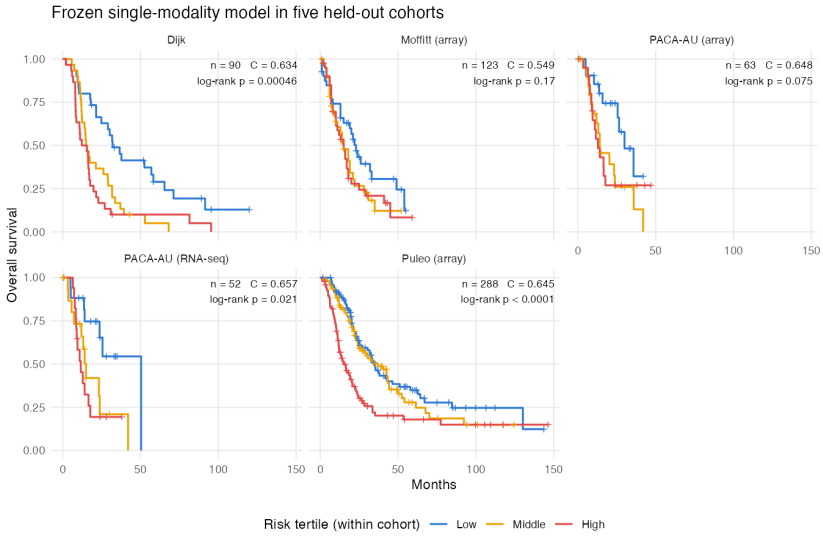


Figure 1: Overall survival by within-cohort tertile of the frozen single-modality risk score ($-\eta$) in the five external cohorts. Annotations give cohort size, concordance and the log-rank p-value. Source: S36, S49.

3.3 Starting rank and factor classification

At starting rank 7 the established fit has three dead factors, two genomics-only programs (5 and 6), and two survival-active programs (3 and 7); the retained programs explain 1.9%–3.0% of the training variance each (S16, S58). Table 4 shows the starting-rank sweep. The evidence lower bound

and BIC both prefer $K_{\text{init}} = 3$; the joint log-likelihood keeps rising to $K_{\text{init}} = 15$; cross-validated survival log partial likelihood is highest at 19; and external concordance is below 0.60 up to rank 6, then lies between 0.626 and 0.628 at every starting rank from 7 to 20 except 11 (0.600) and 13 (0.539). The number of survival-active programs is 1 at starting ranks 2–4 and 2 or 3 from rank 5 upward. Starting rank 7 is the smallest rank on the external-concordance plateau and the rank selected by training cross-validation (Section 1.4). The dated record attributes the dips at ranks 11 and 13 to single-start fitting failures, and an earlier explanation of the gap between ELBO and external performance in terms of reconstruction remains a hypothesis that was not verified (S01).

Table 4: Single-modality starting-rank sweep (selected rows of $K_{\text{init}} = 2, \ldots, 20$; all fits converged). ELBO is the monitoring objective at convergence; BIC uses $K_{\text{init}}(n + p + 1)$ parameters; CV log PL is five-fold cross-validated Cox partial log-likelihood per event; external C uses the five held-out cohorts and was part of the evidence for the recommendation. Source: S14.

Starting rank	Survival-active	Genomics-only	ELBO	BIC	CV log PL per event	Mean external C
3	1	2	−806,783	1,575,641	−3.222	0.594
5	3	1	−808,550	1,581,604	−3.205	0.596
7	2	2	−810,687	1,614,080	−3.230	0.627
9	2	3	−810,568	1,623,226	−3.212	0.627
10	2	3	−814,091	1,636,335	−3.194	0.628
12	2	3	−820,757	1,662,571	−3.176	0.627
15	2	3	−826,829	1,697,684	−3.195	0.627
20	2	3	−847,947	1,767,490	−3.183	0.627

At starting rank 7 the classification is unchanged (two survival-active, two genomics-only) for coefficient thresholds from 10^{-4} to 10^{-2} combined with PVE thresholds from 0.1% to 2%. A PVE threshold of 5% removes both genomics-only programs; a coefficient threshold of 0 counts four programs as survival-active because two coefficients are of order 10^{-10}; and a threshold of 0.05 counts none (S15).

3.4 Multimodal simulation

With pruning, the final number of programs depends much less on the starting rank (Figure 2). The original model has no removal step, so its final rank equals the starting rank in all 162 fits. At moderate noise the evaluated framework returns a mean of 3.2–3.3 programs in the two scenarios with high-variance prognostic programs and 2.5 in the low-variance scenario, averaged over starting ranks 3–15, and the means are close to constant from starting rank 4 upward. At the September noise level it returns fewer than three programs on average (1.7–2.0), and the final rank rises with the starting rank, from 1 at starting rank 3 to 2.3–3.3 at ranks 10 and 15; it is therefore not independent of the starting rank in that setting. All 162 fits of the evaluated framework met the convergence criterion, against 27 of 162 for the original model (S26).

Table 5 reports recovery by scenario and noise. At moderate noise, the framework recovers all three programs in the two-prognostic scenario and 2.96 of 3 in the adverse-and-protective scenario, finds 87% of prognostic programs with the correct sign, and has validation C of 0.79–0.80. In the low-variance scenario at moderate noise it recovers about two programs, finds the low-variance prognostic program in none of the 27 fits (recall 0), and classifies 1.26 other programs per fit as

survival-active, so survival is attributed to high-variance programs; validation C is 0.63. At the September noise level, recovery is lower (0.93–1.33 programs), but the low-variance prognostic program is found in 52% of fits (48% with the correct sign). That result comes from a setting in which every true program is weak, so it does not show that the method recovers a low-variance program alongside strong ones. The variance-corrected survival term gave nearly identical results (largest difference in any summary in -mm-sim, 0.04). Each cell averages 27 fits from only three simulated datasets per setting, which limits precision.

Table 5: Multimodal simulation, evaluated pruned framework; values in parentheses are the original September model on the same datasets. Each cell averages 27 fits (starting ranks 3–10 and 15, three seeds). "Prognostic found, correct sign" is the fraction of true prognostic programs matched at $|r| \geq 0.7$ by a survival-active estimate with the correct coefficient sign. "False survival-active" counts survival-active estimates not matched to a true prognostic program. Validation C uses frozen orientation on 120 validation patients. Source: S26, S27.

Noise	Scenario	Final rank	Programs recovered (of 3)	Prognostic found, correct sign	False survival-active	Validation C	Converged
Moderate	Two prognostic	3.26	3.00 (2.74)	0.87	0.67	0.80	1.00 (0.07)
Moderate	Adverse + protective	3.22	2.96 (2.78)	0.87	0.59	0.79	1.00 (0.37)
Moderate	Low-variance prognostic	2.48	1.96 (2.00)	0.00	1.26	0.63	1.00 (0.19)
September	Two prognostic	1.74	1.19 (0.11)	0.57	0.44	0.79	1.00 (0.07)
September	Adverse + protective	1.96	1.33 (0.59)	0.67	0.52	0.80	1.00 (0.22)
September	Low-variance prognostic	1.96	0.93 (0.15)	0.48	1.15	0.81	1.00 (0.07)

3.5 Multimodal real-data validation

Table 6 compares models on the same 50 primary-PDAC ICGC donors and the 67-donor sensitivity set at starting rank 7. The joint expression-and-methylation fit has $C = 0.685$ (0.579, 0.779) and the expression-only fit of the same framework $C = 0.676$ (0.569, 0.770). Both fits stopped at the 500-sweep cap without meeting the convergence criterion. The difference of 0.010 is small relative to either interval and was not tested; similar observed concordance is not evidence of equivalence. On all 67 donors the order reverses (0.639 and 0.653). The two-step comparators have concordance below 0.53, and the September version of the joint model 0.453.

The October progress chapter (S39) reports that the established single-modality model has $C = 0.685$ on the same 50 donors. That value is hard-coded in the plotting script, and no saved score file or computation was found for it during drafting, so it is not used here as a verified result; it appears in the evidence-gap table (Table 11). The established model was in any case trained on different data (TCGA and CPTAC, per-platform standardized, 2,064 genes), and the 50 donors overlap the PACA-AU RNA-seq cohort on which it was already evaluated.

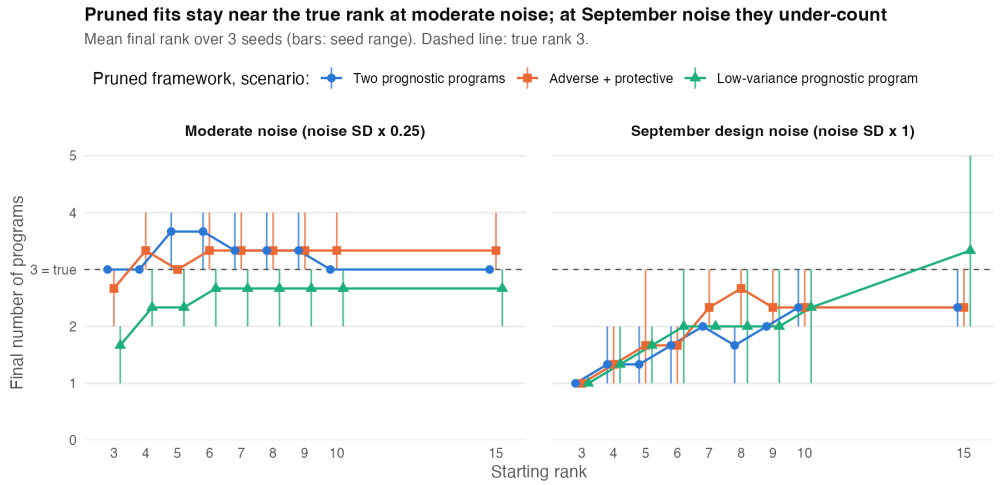


Figure 2: Final number of programs after pruning, against starting rank, in the multimodal simulation (true rank 3, dashed line). Lines show the evaluated pruned framework in each scenario: points are means over three seeds and vertical bars span the seeds; points are offset horizontally for legibility. The original September model is not drawn: it has no removal step, so its final rank equals the starting rank in every fit. The variance-corrected variant is omitted because no summary in Table 5 differs from the plotted framework by more than 0.04. Noise is a multiplier on every feature's noise standard deviation. Sources: S26, S62; drawn for this report by `figures/make_pruning_rank_figure.R` from the saved sweep, replacing the progress-chapter figure.

Table 6: Matched-donor comparisons at starting rank 7, frozen orientation, bootstrap 95% intervals. Selectors: `K_init == 7`; the joint and expression-only rows are `method == joint_multimodal_yfb` with `config == pruned` and `config == pruned_expression_only`; the older fitter is `method == expression_only_yfb`; the September model is `config == original`. Source: S28.

Model (training: TCGA, $n = 144$)	Data	ICGC primary PDAC ($n = 50$, 30 events)	ICGC all donors ($n = 67$, 40 events)	TCGA training	Converged
Evaluated framework, joint	Expression + methylation	0.685 (0.579, 0.779)	0.639 (0.547, 0.734)	0.638	No (500 sweeps)
Evaluated framework, expression only	Expression	0.676 (0.569, 0.770)	0.653 (0.568, 0.744)	0.644	No
September joint model	Expression + methylation	0.453 (0.319, 0.599)	0.442 (0.335, 0.562)	0.597	No
Older isolated single-modality fitter	Expression	0.491 (0.364, 0.633)	0.480 (0.369, 0.581)	0.594	–
Two-step factorization, then Cox	Expression	0.525 (0.399, 0.651)	0.527 (0.425, 0.641)	0.670	–

Model (training: TCGA, $n = 144$)	Data	ICGC primary PDAC ($n = 50$, 30 events)	ICGC all donors ($n = 67$, 40 events)	TCGA training	Converged
Two-step factorization, then Cox	Expression + methylation	0.424 (0.286, 0.567)	0.418 (0.310, 0.531)	0.579	–

Table 7 shows every starting rank. No factor was removed at any starting rank: the final rank equals the starting rank in all nine fits (S28). Only the fits at ranks 3 and 4 met the convergence criterion; the other seven stopped at the cap, and the removal check ran on those unconverged fits (Section 1.6). Concordance on the 50 primary donors ranges from 0.500 at rank 5, where no program is survival-active and every patient receives the same score, to 0.685 at rank 7; the median over the nine ranks is 0.572. The number of survival-active programs ranges from 0 to 11 and rises with the starting rank, while concordance does not. Rank 7 was chosen in advance as the single-modality default, but it is also the rank with the highest validation concordance in this sweep, so the headline rank-7 results are favourable relative to the other ranks.

Table 7: Joint multimodal fits across starting ranks (`config == pruned`, `method == joint_multimodal_yfb`). Source: S28.

Starting rank	3	4	5	6	7	8	9	10	15
Final rank	3	4	5	6	7	8	9	10	15
Survival-active programs	0	1	0	0	1	2	1	3	11
Converged	Yes	Yes	No	No	No	No	No	No	No
ICGC primary C	0.557	0.574	0.500	0.546	0.685	0.589	0.572	0.650	0.547
ICGC all-donor C	0.499	0.502	0.500	0.522	0.639	0.545	0.547	0.610	0.532

The October progress chapter (S39) attributes the absence of pruning to each program's molecular reconstruction gain being far larger than its complexity cost, with survival changing the evidence by at most a few units. That per-program decomposition was not traced to a saved output during drafting and is not quantified here. The plug-in survival term contributes on the scale of a 75-event partial likelihood, while the reconstruction term sums over $144 \times 13{,}000$ molecular values, so in the evaluated objective survival can have little influence on which programs are retained.

4 Biological findings

4.1 Single-modality programs

The established fit retains four programs (Table 8). Each label below is tied to the analyses of Table 14 that support it. Two general points apply. First, the loadings of these programs were estimated without survival ($\alpha_F = 0$), so the biology describes the dominant expression structure of the training data, which survival then weights; it is not evidence of survival-guided discovery. Second, the normalized enrichment scores are modest (1.05–1.29 for the sets cited), the fgsea test is one-sided toward high loadings, and multiplicity is controlled only within each program and collection.

Program 7: adverse, basal-like. Its highest loadings combine glycolytic genes (GAPDH, ENO1, TPI1, PGK1, LDHA) with integrin and receptor-tyrosine-kinase genes (ITGA3, ITGB4, MET, CDCP1). Its enriched sets include DeSurv's basal-like list (padj 0.0006), KEGG growth-factor RTK–PI3K and RAS–ERK pathways (padj 0.001) and Bailey squamous (padj 0.03) (S22). Training activities correlate with PurIST basal-like probability ($\rho = 0.58$; S23); its top genes overlap only DeSurv's basal-like list (Jaccard 0.18; S35); the basal-like signature has HR above 1 in all five external cohorts (1.30–2.40 per unit of mean score, $p < 0.05$ in two; S24); and projection tertiles separate external survival (Figure 3). It corresponds to an unsupervised factor of the same data ($r = 0.69$, next best 0.33; S45).

Program 3: protective, classical. The top genes are epithelial and secretory (CAPN5, LPCAT4, MLPH, SLC45A3, TJP3, GALNT6). The classical label rests mainly on subtype concordance ($\rho = -0.58$) and the classical signature's HR below 1 in all five cohorts (0.42–0.88, $p < 0.05$ in three). The enrichment is less specific: DeSurv's classical and basal-like lists are both enriched (padj 0.017 each), with Moffitt classical and three integrin–FAK sets, and the top genes overlap both lists (Jaccard 0.13 and 0.10). Its unsupervised correspondence is less distinct ($r = -0.72$, next best 0.60).

Program 5: genomics-only, desmoplastic stroma. Collagen and fibroblast genes dominate (COL1A2, COL3A1, COL5A2, THBS2, CDH11, SPARC); its twelve sets are led by Hallmark EMT (padj 0.0005), DeSurv's stromal-immune list and extracellular-matrix organization; its top genes overlap only the stromal-immune list (Jaccard 0.18); it leans weakly basal ($\rho = 0.30$); and its EMT signature is not consistently prognostic externally (HR 0.73–1.50, $p < 0.05$ in one cohort; S40).

Program 6: genomics-only, immune infiltrate with a vascular component. Lymphoid genes (IKZF1, AKNA, IL10RA, RCSD1, PREX1) are mixed with vascular and basement-membrane genes (APOLD1, COL14A1, TNXB, FBLN5, SLIT3, SPARCL1), so the label keeps both components, following the August 21 report (S21). Most of its 40 sets (S61) are immune GO terms, though the two lowest padj values are DeSurv's classical and stromal-immune lists. It is uncorrelated with the PurIST axis ($\rho = -0.03$) but differs across PurIST calls (Kruskal–Wallis $p = 0.007$), a non-monotonic relation. Scored on its own immune set after the post hoc C4 correction, it has no consistent external association (HR 0.53–1.39, none with $p < 0.05$; S40).

Together, the four programs map onto DeSurv's three published factors, with DeSurv's combined stromal-immune factor divided between a stromal and an immune program. This is gene-list correspondence within a DeSurv-aligned gene universe, not an independent replication of DeSurv and not a predictive comparison with it. The August 21 report also noted that the three dead programs have no enriched sets; their posterior-mean loadings are exactly zero in the saved fit, so

that observation follows from the fit and does not corroborate the classification independently.

Table 8: Retained single-modality programs at starting rank 7. Risk-scale coefficients are $-\tilde{\tilde{\beta}}_k$ of the saved fit, per unit of normalized projection; they agree in sign with the marginal direction labels. Sources: S16, S21–S24, S35, S40, S45, S58.

Program	Class	Label	Risk-scale coefficient	Supporting analyses (rows of Table 14)
7	Survival-active	Adverse; basal-like	+0.040	fgsea (DeSurv basal-like, RTK–PI3K/ERK); $\rho = 0.58$ with PurIST; overlap with DeSurv basal-like only; external signature HR > 1 in 5/5 cohorts; tertile curves
3	Survival-active	Protective; classical	−0.012	$\rho = -0.58$ with PurIST; external signature HR < 1 in 5/5 cohorts; tertile curves; fgsea and overlap include both classical and basal-like lists
5	Genomics-only	Desmoplastic stroma	≈ 0	fgsea (EMT, ECM); overlap with DeSurv stromal-immune only; weak basal lean; external signature not consistently prognostic
6	Genomics-only	Immune infiltrate with a vascular component	≈ 0	fgsea (immune GO terms; DeSurv lists); overlap with stromal-immune and classical lists; non-monotonic subtype relation; corrected external control null

Direction labels. Adverse and protective labels follow the marginal association of each program's projection with survival (S01). The pathway report read the stored coefficients ($\tilde{\beta}_7 = -0.040$, $\tilde{\beta}_3 = +0.012$) as risk-scale values and described a sign disagreement with the marginal directions. Because the saved fit's η is a good-prognosis score (Section 1.3.1), the risk-scale coefficients are +0.040 for Program 7 and −0.012 for Program 3, which agree in sign with the marginal labels. These coefficients are conditional on the other programs and are not interchangeable with the marginal associations.

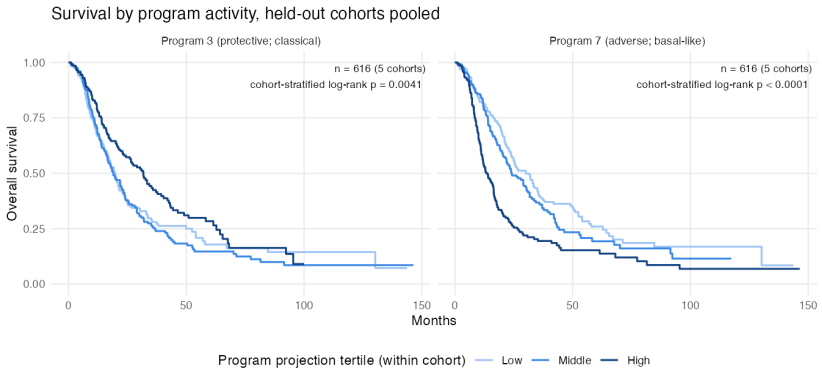


Figure 3: Overall survival by within-cohort tertile of the frozen Program 3 and Program 7 projection scores, pooled over the five external cohorts ($n = 616$), with cohort-stratified log-rank tests annotated. These are projection scores, not training activities $E(L)$ or signature means. Source: S49; figure from the October 2 progress chapter.

4.2 Multimodal programs

The starting-rank-7 joint fit has seven programs (Table 9). Labels are provisional: they come from a manual reading of each program's eight highest positive and six most negative expression loadings, together with the saved loading correlations to three references. No enrichment test and no CpG annotation was performed for these programs. Because the saved correlations are absolute values, the tables cannot show whether a matched pair points in the same direction, and best matches chosen independently for each program need not form a one-to-one alignment; for example, both the immune and stromal programs have their best established-model match in Program 6.

Table 9: Multimodal programs at starting rank 7. Label basis: manual reading of top signed expression genes, plus best absolute loading correlation with the established single-modality programs (D4; 1,781 shared genes), an unsupervised `flash` fit of TCGA expression (14 factors; 3,000 genes), and the expression-only fit of the same framework. Shares are within-modality rank-one shares of fitted signal. Class: reconstruction-active only, or both reconstruction- and survival-active. HRs are univariable, per one within-cohort SD of the joint projection, and unadjusted; 14 tests. ICGC is the 50-donor primary-PDAC set. Sources: S31, S32, S43, S44.

Program	Provisional label	Label basis: top signed expression genes; correspondence $\|r\|$	Share of expr / meth signal	Class	HR per SD, TCGA (p)	HR per SD, ICGC (p)
1	Classical / gastric	+ REG4, TFF1, CLDN18, CEACAM5; − CXCL12, ADAM12, CILP. D4 P3 0.79; flash k5 0.79; expr-only 0.87	23.5% / 4.8%	Reconstruction	1.07 (0.58)	0.79 (0.30)
2	Immune	+ CCL19, CORO1A, PTPRC, CCL21, LTB; − FOXA2, H1F0. D4 P6 0.52; flash k9 0.63; expr-only 0.67	5.1% / 20.9%	Reconstruction	0.85 (0.17)	0.87 (0.54)
3	Mixed (no label)	+ MIF, TNFRSF6B, CCDC85B, SYT8; − IL6ST, HIPK3, EDIL3. D4 best 0.36; flash k3 0.78; expr-only 0.79	31.5% / 0.3%	Reconstruction	1.00 (0.98)	1.48 (0.051)
4	Basal-like vs classical	+ S100A2, MUC16, KRT5, KRT6A, KRT17; − CLDN18, LYZ, REG4. D4 P7 0.68; flash k8 0.70; expr-only 0.88	6.2% / 4.5%	Both	1.34 (0.0055)	1.99 (0.00033)
5	Normal vs activated stroma	+ ADH1B, C7, CHRDL1, PRUNE2; − COL11A1, MMP11, MSLN, COL17A1. D4 P6 0.48; flash k2 0.60; expr-only 0.52	4.8% / 1.2%	Reconstruction	0.75 (0.043)	0.83 (0.46)
6	Exocrine / acinar	+ CTRB2, CPB1, CPA1, PNLIP, PRSS1; − LRRC15, COL11A1, GREM1. D4 best 0.38; flash k6 0.87; expr-only 0.91	28.9% / 9.3%	Reconstruction	0.92 (0.53)	1.17 (0.37)
7	Methylation-only	No expression loading; no expression correspondence	0% / 58.9%	Reconstruction	1.11 (0.36)	0.87 (0.44)

The one survival-active program (4) is the basal-like versus classical axis. Its marginal hazard ratio per SD of the joint projection is 1.99 in ICGC ($p = 0.00033$) and 1.34 in TCGA ($p = 0.0055$). It corresponds to established Program 7 ($|r| = 0.68$), and the classical/gastric program (1) corresponds to established Program 3 ($|r| = 0.79$), although Program 1 is not survival-active here. Of the 14 marginal tests, three have $p < 0.05$ (Program 4 in both cohorts and Program 5 in TCGA); no correction for multiplicity was applied.

Methylation contributes in two ways (Figure 4). A methylation-only program carries 58.9% of the fitted methylation signal and has no expression loading. The immune program carries 20.9% of the methylation signal while being defined on the expression side by immune genes. Neither is survival-active, and neither has a marginal association with $p < 0.05$. The basal-like program carries 4.5% of the methylation signal. These shares describe the fitted rank-one reconstruction within each modality. They are not additive proportions of variance, because cross-terms between programs are excluded, and they say nothing about each program's contribution to survival prediction.

The current evidence therefore supports expression-defined program labels and the existence of methylation-dominant structure in the screened CpGs. It does not establish what that structure is. The 10,000 CpGs were chosen by variance, and high-variance CpGs often track cell composition, so the immune and methylation-only programs may reflect composition or technical variation; this is an explanation to test, not a finding.

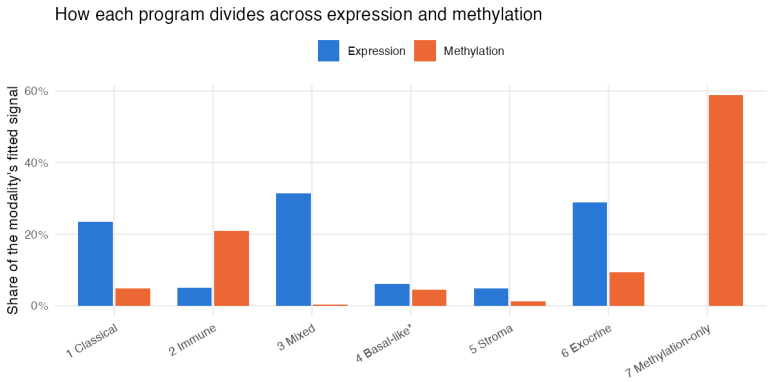


Figure 4: Share of each modality's fitted rank-one signal carried by each program of the starting-rank-7 joint fit. The asterisk marks the survival-active program. Shares are not additive variance explained and not survival contributions. Source: S31; figure from the October 2 progress chapter.

Table 10: Multimodal characterization completed to date and characterization that remains to be done. Sources: S31, S32, S43, S44, S47, S38.

Completed multimodal characterization	Proposed characterization (not completed evidence)
Signed top-expression-gene lists and manual provisional labels	Sign-aware, two-sided enrichment of expression loadings; the existing signed enrichment of unsupervised reference factors (S47) shows an alignment convention, not a completed multimodal analysis
Within-modality rank-one fitted-signal shares	CpG annotation and genomic-context enrichment
Absolute loading correlations with expression-only, established D4, and unsupervised `flash` references	Checks of cell-composition and technical explanations; direction-aware correspondence if later computed
Fourteen unadjusted, within-cohort standardized joint-projection Cox associations	Characterization and validation of the final programs after the method work

5 Interpretation and remaining evidence

5.1 What the current evidence supports

The single-modality model yields a few interpretable programs and a risk score that transfers across platforms, but its advantage over a tuned two-step pipeline is small and marginal, and with $\alpha_F = 0$ that advantage comes from how programs are fitted and coefficients shrunk, not from survival-guided discovery. Its programs match the known basal-like/classical and stromal/immune axes of PDAC. The multimodal framework fixes the September model's failure to remove factors and works as designed in simulation when the true programs explain enough variance; on real data it has not shown that methylation improves prediction, nothing is pruned, and most fits are unconverged. Neither model recovers a planted low-variance prognostic program alongside strong ones, which is the case a supervised factorization exists for.

5.2 Evidence gaps by claim

Table 11 lists the six claims a manuscript would need to make, the evidence that exists, the conclusion it justifies now, and what is missing. It records the present state of the evidence; it does not imply that the missing analyses were run for this report.

Table 11: Claim-centered evidence gaps. File names are abbreviated; the full repository path for each source ID is recorded in `source_inventory.csv`.

Claim	Existing evidence	Source ID, path and selector	Currently justified conclusion	Missing evidence
Value over EBMF or NMF alone	Simulated recovery of shared and specific programs; established Programs 3 and 7 correspond to unsupervised EBMF factors; their pathway results partly agree; multimodal programs correspond to an unsupervised `flash` fit	S18 (`multicohort_sim_results.csv`, scenario × K_init × arm); S19 (`top2_match_diagnostic.csv`); S45 (`T5_ebmf_factor_correspondence.csv`, program 3, 7); S46 (`T6_ebmf_pathway_concordance.csv`: 5 of 13 sets concordant); S32 (`real_factor_comparison.csv`, `reference == flash`)	The programs largely coincide with unsupervised structure; recovery differences in simulation reflect the fitting procedure, not supervision, under $\alpha_F = 0$	A survival comparison with unsupervised factorization needs an explicit scoring rule (for example, factor scores entered into a Cox model). No NMF comparison was found in the reviewed comparator sources
Value over two-step EBMF → Cox	Principal concatenated difference 0.026 (0.0002, 0.0498) against 40 factors with LASSO; rank-matched difference 0.011 (−0.013, 0.033); older unpenalized difference 0.042; simulation two-step arm with unpenalized Cox	S10 (`ebmf_cox_regularized_results.csv`, `k_tag == "k40"`); S11 (`yfb_vs_ebmf_k40_pooled_ci.csv`); S12 (`yfb_vs_ebmf_k7_matched_ci.csv`, `cohort == "POOLED"`); S13 (`external_paired_diff_ci.csv`, pooled row); S18	A small, marginal advantage over the principal comparator in this benchmark; no advantage shown over the rank-matched comparator	Comparison of the final method with the same comparators in simulation and real data, with replicated starts

Claim	Existing evidence	Source ID, path and selector	Currently justified conclusion	Missing evidence
Value over DeSurv	Gene-list overlap of the four programs with DeSurv's three factors; our models fitted with DeSurv-aligned preprocessing	S35 (`T4_sbmf_desurv_overlap.csv`, program × desurv_factor); S48 (`run_desurv_comparison.R`, configurations D1–D5)	Correspondence of gene lists within a DeSurv-aligned gene universe. No predictive comparison with DeSurv exists in the reviewed sources	A matched head-to-head with DeSurv on the same training and validation data. The directory name `desurv_comparison` does not indicate a DeSurv fit
Rank selection accounting for survival and reconstruction	Single-modality sweep with disagreeing criteria and an external-concordance plateau from 7; threshold sensitivity; multimodal simulation final ranks; real multimodal fits never pruned; removal criterion with plug-in survival	S14 (`k_init_sweep_results.csv`); S55 (`kcv_yfb_desurv_corrected.csv`); S15 (`threshold_grid_K7.csv`); S26, S27 (`pruning_comparison*.csv`, `variant == "pruned"`); S28 (`real_sweep_summary.csv`, `K_final`); S01; S08	Starting rank 7 is supported by training cross-validation and by external performance, which was used in the recommendation; criteria disagree; the multimodal removal step works in simulation but not on real data	A selection rule that combines survival and reconstruction, validated with replicated simulations and without validation outcomes. The dated explanation of the ELBO-versus-performance gap is unverified, and the October per-program evidence decomposition was not traced to a saved output
Multimodal added value	Joint and expression-only fits of the same framework on the same 50 donors at starting rank 7; seven programs with modality shares and correspondence	S28 (`K_init == 7`, `cohort == "icgc_primary"`, `method == "joint_multimodal_yfb"`, `config == "pruned"` or `"pruned_expression_only"`); S31 (`real_program_table_K07.csv`); S32	Similar observed concordance (0.685 and 0.676), both unconverged; methylation adds methylation-dominant structure	A test of improved prediction; converged fits; robustness to starting rank, CpG selection and feature counts. The established single-modality model's C on the same donors (0.685 in the October chapter) is hard-coded in `plot_real_cindex_by_kinit.R` and has no saved score source
Biological understanding	Single-modality enrichment, subtype concordance, overlap and external signature checks; multimodal top genes, shares, absolute correspondence and 14 unadjusted HRs	S21–S24, S35, S36, S40–S44, S49, S51–S54	Descriptive program labels consistent with known PDAC axes; checks share inputs, include a post hoc correction and within-training concordance, and use unadjusted tests and unsigned matching	Mechanism; characterization of the final programs; signed enrichment of multimodal loadings; CpG annotation; assessment of the characterization methods themselves

5.3 Limitations

The external cohorts differ in platform, size and event rate, and two of them come from the same ICGC project; the matched multimodal validation set overlaps one of them. The multimodal real-data results rest on one training cohort of 144 patients with 75 deaths and a 50-donor validation set, so concordance intervals are wide (about ± 0.10). Rank 7 is favourable among the multimodal starting ranks evaluated. The multimodal simulation uses three datasets per setting. Feature screening keeps 3,000 genes and the 10,000 highest-variance CpGs, about 3% of those available, and the effect of that choice is unknown. The single-modality simulation reports outcome-oriented concordance. The saved historical single-modality fit predates the September orientation correction and has no stored coefficient uncertainty. The biological characterization methods were chosen for availability and have not been assessed.

5.4 Continuing work

The configuration with $\alpha_F = 0$ is not the intended final method: a supervised factorization should be able to find a prognostic program that explains little molecular variance, and the evaluated configuration cannot, because survival never enters the loading update. The planned work, recorded in the October 2 follow-up plan and its October 13 sequencing (S38), has five parts. These are plans; no result from them is reported here.

1. **Survival supervision of the loadings.** Diagnose why survival has little effect even when it enters the loading update: its contribution is multiplied by the current coefficient, so a direction no program already points at receives no survival information; its precision is small relative to the reconstruction term; and in the single-modality model the loading scale is not fixed. The work adds the multimodal scale canonicalization to the single-modality model, removes α_F as a separate setting, tests a supervised initialization, and compares a coefficient prior with an explicit survival weight. Success is defined in advance as recovery of a planted low-variance prognostic program in a large majority of simulation seeds, without instability and without loss of external concordance.
2. **Notation and prediction.** Settle the definition of the normalized projection and the coefficient units once the scale canonicalization is in place, and carry the chosen supervision method into the multimodal model.
3. **Rank selection.** Compare an evidence criterion with weighted survival against selection by cross-validated concordance plus reconstruction, with replicated simulation panels and a one-standard-error rule, for both models.
4. **Robustness and comparators.** Repeat fits from jittered starts, compare loading prior families, vary the number of genes and CpGs, and run a comparator matrix in simulation and on real data that includes unsupervised factorization, the two-step pipeline, and DeSurv.
5. **Biology of the final programs.** Recompute risk scores, survival curves and enrichment for the final programs, and add signed enrichment and CpG annotation for the multimodal programs.

References

Young, Amber M., Alisa Yurovsky, Xianlu Laura Peng, Jen Jen Yeh, Didong Li, and Naim U. Rashid. 2026. “Survival-Guided Matrix Factorization Identifies Reproducible Prognostic Programs in

Pancreatic Cancer." Unpublished manuscript.

Appendix A. Single-modality derivations

This appendix reconciles the existing single-modality derivations (S06, S07) with the evaluated D4 fit and the current implementation in `code/fit_cox_on_yf.R` and its update files (S04). It derives the updates the evaluated algorithm uses, states where the implementation departs from the joint mathematical model, and lists the approximations. Equation numbers continue from the main text.

A.1 Model and variational family

The joint model combines the reconstruction Equation 1 with a Cox model for (t_i, δ_i) whose linear predictor is $\eta_i = \tilde{z}_i^\top \tilde{\beta}$, where $\tilde{z}_i = D^{-1}\bar{F}^\top y_i$ is the normalized projection of Equation 2. Priors are $L_{ik} \sim g_{L,k}$ and $F_{jk} \sim g_{F,k}$ (point-exponential) and $\tilde{\beta}_k \sim g_{\beta,k} = N(0, \sigma_k^2)$. The variational family is fully factorized,

$$q(L, F, \tilde{\beta}) = \prod_{i,k} q(L_{ik}) \prod_{j,k} q(F_{jk}) \prod_k q(\tilde{\beta}_k),$$

with τ and the prior parameters treated as point estimates. Write $\bar{L}_{ik} = E_q L_{ik}$ and $\overline{L^2}_{ik} = E_q L_{ik}^2$, and similarly for F and $\tilde{\beta}$.

Two features of the predictor matter for every derivation below. First, $\tilde{z}_i$ is formed from the posterior mean $\bar{F}$, not from F itself, and D is computed from $\bar{F}$ at the start of each outer iteration. The projection is therefore a fixed covariate within an iteration: it has no posterior variance in the coefficient update. Second, L does not appear in η.

A.2 Cox working approximation

Let $\hat{\eta}$ be the predictor at the start of an iteration, $\hat{\theta}_i = \exp(\hat{\eta}_i)$, and $\hat{H}_0$ the Breslow cumulative baseline hazard with ties sharing a risk-set denominator. The score and the diagonal of the negative Hessian of $\log \mathrm{PL}$ with respect to η_i are approximated by

$$u_i = \delta_i - \hat{\theta}_i \hat{H}_0(t_i), \qquad w_i = \hat{\theta}_i \hat{H}_0(t_i), \tag{7}$$

with w_i floored at 10^{-6}. A second-order expansion with this diagonal curvature gives

$$\log \mathrm{PL}(\eta) \approx c - \tfrac{1}{2} \sum_i w_i (z_i - \eta_i)^2, \qquad z_i = \hat{\eta}_i + u_i / w_i. \tag{8}$$

The exact Hessian of the partial likelihood is not diagonal; dropping its off-diagonal terms is the first approximation. When `strata_id` is supplied, risk sets and $\hat{H}_0$ are formed within strata; the evaluated fit has no strata.

A.3 EBNM handoff

Every coordinate update below reduces to the same form. If the expected log objective, as a function of one scalar parameter θ, is $B\theta - \frac{1}{2}A\theta^2 + \text{const}$ with $A > 0$, it equals the log-likelihood of a single Gaussian observation

$$x = B/A, \qquad x \mid \theta \sim N(\theta, s^2), \qquad s = A^{-1/2}. \tag{9}$$

The posterior under the prior g is the empirical-Bayes normal-means posterior for (x, s), computed by `ebnm`, which also estimates g by marginal likelihood when g is shared across a vector of observations.

The KL term needed for the objective follows from the identity

$$\mathrm{KL}(q \,\|\, g) = E_q\big[\log N(x; \theta, s^2)\big] - \log p(x), \tag{10}$$

where $p(x)$ is the marginal likelihood returned by `ebnm`.

A.4 Coefficient update

Isolate factor k: $\eta_i = \tilde{Z}_{ik}\tilde{\beta}_k + \eta_{i,-k}$, with $\eta_{i,-k} = \sum_{l \neq k} \tilde{Z}_{il}\bar{\tilde{\beta}}_l$. Substituting into Equation 8 and taking the expectation over $q(\tilde{\beta}_k)$ with $\tilde{Z}$ fixed, the terms in $\tilde{\beta}_k$ are

$$\tilde{\beta}_k \sum_i w_i \tilde{Z}_{ik}(z_i - \eta_{i,-k}) - \tfrac{1}{2}\tilde{\beta}_k^2 \sum_i w_i \tilde{Z}_{ik}^2.$$

The implementation multiplies the survival term by the weight α:

$$A_{\beta,k} = \alpha \sum_i w_i \tilde{Z}_{ik}^2, \qquad B_{\beta,k} = \alpha \sum_i w_i \tilde{Z}_{ik}(z_i - \eta_{i,-k}). \tag{11}$$

The pseudo-observation B/A does not depend on α; its standard error is inflated by $\alpha^{-1/2}$ ($\sqrt{2}$ at the evaluated $\alpha = 0.5$). The prior is normal with variance estimated by `ebnm` from this single pseudo-observation, which gives $\hat{\sigma}_k^2 = \max(0, x^2 - s^2)$: a coefficient whose pseudo-observation is smaller than its standard error is set exactly to zero. With $\tilde{Z}$ treated as fixed, Equation 11 uses $\tilde{Z}_{ik}^2$ where the joint model would use $E(\tilde{Z}_{ik}^2)$; posterior uncertainty in F does not reach the coefficient.

A.5 Activity update

Only the reconstruction term contains L. With the factor-k residual $R^{(-k)} = Y - \sum_{l \neq k} \bar{L}_{\cdot l}\bar{F}_{\cdot l}^{\top}$, the expected log-likelihood of Equation 1 contributes, for patient i,

$$L_{ik} \sum_j \tau_j R_{ij}^{(-k)} \bar{F}_{jk} - \tfrac{1}{2} L_{ik}^2 \sum_j \tau_j \overline{F^2}_{jk},$$

so

$$A_{L,k} = \sum_j \tau_j \overline{F^2}_{jk}, \qquad B_{L,ik} = \sum_j \tau_j R_{ij}^{(-k)} \bar{F}_{jk}, \tag{12}$$

with one shared precision $A_{L,k}$ across patients and a point-exponential prior fitted by `ebnm` over the n pseudo-observations. This is the standard empirical-Bayes matrix factorization update; it holds in every configuration of the single-modality model because η does not depend on L.

A.6 Loading update

Reconstruction contribution. By the same argument as Equation 12 with the roles of L and F exchanged,

$$A_{F,jk}^{\mathrm{gen}} = \tau_j \sum_i \overline{L^2}_{ik}, \qquad B_{F,jk}^{\mathrm{gen}} = \tau_j \sum_i R_{ij}^{(-k)} \bar{L}_{ik}. \tag{13}$$

Survival contribution in the joint model. Because η depends on F, the joint model also has a survival term in the loading update. Treating the normalizing constant d_k as fixed and writing

$\tilde{Z}_{ik} = d_k^{-1} \sum_j y_{ij} F_{jk}$, the terms of Equation 8 in F_{jk} are, up to factors of d_k,

$$F_{jk}\, E(\tilde{\beta}_k) \sum_i w_i y_{ij} (z_i - \eta_{i,-k}) - \tfrac{1}{2} F_{jk}^2 E(\tilde{\beta}_k^2) \sum_i w_i y_{ij}^2 - F_{jk} E(\tilde{\beta}_k^2) \sum_i w_i y_{ij} \tilde{Z}_{ik}^{(-j)},$$

where the last term couples feature j to the other features through the projection excluding j. A full treatment would also differentiate d_k with respect to F_{jk}.

Implemented formula. `update_F_surv_YFB_k()` combines the two sources with weights $1 - \alpha_F$ and α_F:

$$A_{F,jk} = (1 - \alpha_F) A_{F,jk}^{\text{gen}} + \alpha_F E(\tilde{\beta}_k^2) \sum_i w_i y_{ij}^2, \quad B_{F,jk} = (1 - \alpha_F) B_{F,jk}^{\text{gen}} + \alpha_F E(\tilde{\beta}_k) \sum_i w_i y_{ij} (z_i - \eta_{i,-k}). \tag{14}$$

This survival branch uses the unnormalized y_{ij}, omits the feature-coupling term and the derivative of d_k, and updates all features of factor k in one vectorized call. The plain Bayesian posterior corresponds to adding both sources with weight 1, which has the same pseudo-observation as $\alpha_F = 0.5$ and twice its precision (S38). **In the evaluated fit** $\alpha_F = 0$, so Equation 14 reduces to Equation 13 and the survival branch is not used. The F update is then the reconstruction update of empirical-Bayes matrix factorization, with the point-exponential prior fitted by `ebnm` over the p pseudo-observations.

A.7 Precision update

Maximizing the expected reconstruction log-likelihood over τ_j gives

$$\tau_j = \frac{n}{\sum_i E_q (y_{ij} - L_i^\top F_j)^2}, \qquad E_q (y_{ij} - L_i^\top F_j)^2 = (y_{ij} - \bar{L}_i^\top \bar{F}_j)^2 + \sum_k (\overline{L^2}_{ik} \overline{F^2}_{jk} - \bar{L}_{ik}^2 \bar{F}_{jk}^2), \tag{15}$$

with a floor that bounds τ_j by 10^8. There is no prior on τ_j.

A.8 Monitoring objective and stopping rule

After each outer iteration the implementation computes

$$J = (1 - \alpha)\, G + \alpha\, p\, S + \sum_k (\kappa_{L,k} + \kappa_{F,k} + \kappa_{\beta,k}), \tag{16}$$

where $G = \sum_j \left[\frac{n}{2} \log \tau_j - \frac{\tau_j}{2} \sum_i E_q (y_{ij} - L_i^\top F_j)^2 \right]$ is the reconstruction log-likelihood without its constant, $S = \log \mathrm{PL}(\hat{\eta}) - \frac{1}{2} \sum_i w_i \mathrm{Var}_q(\eta_i)$ with $\mathrm{Var}_q(\eta_i) = \sum_k \tilde{Z}_{ik}^2 \mathrm{Var}_q(\tilde{\beta}_k)$, and each κ is $-\mathrm{KL}$ computed by Equation 10. The factor p comes from the `per_p` normalization convention, which scales only this monitored quantity. The algorithm stops at the first iteration after the fifth with $|J_t - J_{t-1}| / |J_{t-1}| < 10^{-5}$. Because J weights its components by $1 - \alpha$ and αp, and because $\log \mathrm{PL}$ is evaluated at the iteration's starting predictor, J is a convergence monitor. It is not the evidence lower bound of the joint model, and the updates are not guaranteed to increase it.

A.9 Orientation and prediction

After the loop the implementation checks the training concordance of η once and negates $\bar{\tilde{\beta}}$ if the score is discordant. Before September 4, 2026 this check omitted `reverse = TRUE` in `survival::concordance()`, so it was inverted, and it also replaced $E(\tilde{\beta}^2)$ by $E(\tilde{\beta})^2$ when it negated. The saved D4 fit was produced by that code: its η is a good-prognosis score and its stored coefficient variances are zero. It is scored as risk $= -\eta$ for all held-out cohorts (S01, S49).

For a new patient with data standardized in its own cohort, prediction restricts $\bar{F}$ to the training genes measured in that cohort but keeps the training normalizing constants D, which were computed at the start of the final iteration: $\eta_{\text{new}} = y_{\text{new}}^{\top} \bar{F}_{\mathcal{G}} D^{-1} \bar{\tilde{\beta}}$, where $\mathcal{G}$ is the set of shared genes. Recomputing these projections from the saved fit reproduces the saved held-out scores to within 10^{-8} (S49).

A.10 Mathematical model versus evaluated algorithm

Table 12: Single-modality joint model, implementation, and evaluated fit. Sources: S04, S05, S06, S07, S49, S58.

Component	Joint mathematical model	Implementation (`fit_cox_on_yf()`)	Evaluated D4 fit
Survival term	Exact Cox partial likelihood, expected under q	Diagonal quadratic surrogate Equation 8 at the iteration's starting $\hat{\eta}$	Same; no strata
Predictor	$\eta = \tilde{z}^{\top}\tilde{\beta}$ with $\tilde{z}$ a function of F	$\tilde{Z} = Y\bar{F}D^{-1}$ from posterior means, fixed within an iteration	Same
Coefficient update	Uses $E(\tilde{Z}^2)$ including loading uncertainty	Uses $\tilde{Z}^2$ (plug-in); survival term weighted by α	$\alpha = 0.5$; normal prior with per-factor variance
Activity update	Reconstruction only	Reconstruction only	Same
Loading update	Reconstruction plus survival with feature coupling and $\partial d_k/\partial F$	Weighted sum Equation 14, survival branch without coupling or normalization derivative	$\alpha_F = 0$: reconstruction only
Precision	Maximum likelihood or prior	Maximum likelihood Equation 15	Same
Objective	Evidence lower bound	Weighted monitor Equation 16	Stopped at iteration 20
Orientation	Not part of the model	One post-fit training check	Pre-September check; risk $= -\eta$
Coefficient uncertainty	Posterior variance of $\tilde{\beta}$	$E(\tilde{\beta}^2)$ retained after the September fix	Stored variances are zero (historical code)

The single-modality algorithm therefore performs exact coordinate updates of an empirical-Bayes factorization for L, F and τ, interleaved with a plug-in empirical-Bayes update of the Cox coefficients on fixed projections. The derivation of a survival-informed loading update that is stable is part of the continuing work (Section 5.4).

Appendix B. Multimodal derivations

This appendix derives the updates of the evaluated multimodal framework (`code/fit_multimodal_yfb.R` and `code/multimodal_yfb_updates.R`; S08, S34) and reconciles them with the September base derivation (S33), which is preserved unchanged. The base derivation has nonnegative loadings, no intercept, moment-based prior updates, and no factor removal; the evaluated framework adds a feature intercept, signed point-Laplace loadings, marginal-likelihood prior fits, factor-scale canonicalization, and an evidence-based removal check. Its controls are those of the real-data runner (S29): `intercept = TRUE`, `prior_F = point_laplace` for both modalities, `prior_update = "ebnm"`, `tau_model = "feature"`, `survival_elbo = "plugin"`, `prune = TRUE`, at most 500 outer sweeps of 5 inner sweeps, tolerance 10^{-4}.

B.1 Model

For modalities $m \in \{\text{expr}, \text{meth}\}$, patients i, features j and factors k,

$$y_{mij} = \mu_{mj} + \sum_k l_{ik} f_{mjk} + e_{mij}, \qquad e_{mij} \sim N(0, \tau_{mj}^{-1}), \tag{17}$$

$$l_{ik} \sim g_{L,k} = (1-\pi_{L,k})\delta_0 + \pi_{L,k}\,\text{Exp}(\lambda_{L,k}), \qquad f_{mjk} \sim g_{mk} = (1-\pi_{mk})\delta_0 + \pi_{mk}\,\text{Laplace}(0, \lambda_{mk}^{-1}), \tag{18}$$

and $\beta_k \sim N(0, \sigma_k^2)$. Survival follows a Cox model with

$$\eta_i = \sum_k Z_{ik}\beta_k, \qquad Z_{ik} = \sum_m \sum_j \tilde{y}_{mij} f_{mjk}, \qquad \tilde{y}_{mij} = y_{mij} - \mu_{mj}. \tag{19}$$

The intercept μ_{mj} and precision τ_{mj} are point estimates. Because $\sum_j \mu_{mj} f_{mjk}$ does not depend on i, using y instead of $\tilde{y}$ in Equation 19 changes every η_i by the same constant, which leaves the partial likelihood and the C-index unchanged. Signed loadings are needed because, with an intercept, f_{mjk} measures a deviation from the feature's baseline and can be negative; L stays nonnegative and is read as program activity.

B.2 Variational family and objectives

The variational family is $q = \prod_{i,k} q(l_{ik}) \prod_{m,j,k} q(f_{mjk}) \prod_k q(\beta_k)$. Two objectives are used. The **removal objective** is Equation 6: the reconstruction log-likelihood at the maximum-likelihood $\hat{\tau}$, plus the Breslow log partial likelihood at $E_q(\eta)$, minus the summed coordinate KL divergences computed by Equation 10. The **coordinate updates** use the Cox working surrogate at the predictor $\hat{\eta} = E_q(\eta)$ from the start of the outer sweep,

$$\log \text{PL}(\eta) \approx c + \sum_i (h_i \eta_i - \tfrac{1}{2} w_i \eta_i^2), \qquad h_i = u_i + w_i \hat{\eta}_i, \tag{20}$$

which is Equation 8 rewritten ($h_i = w_i z_i$). The quantities w and h are held fixed for the five inner sweeps of an outer sweep. Its expectation under q is $\sum_i [h_i E(\eta_i) - \tfrac{1}{2} w_i E(\eta_i^2)]$.

B.3 Intercept

At the start of each outer sweep, maximizing the expected reconstruction log-likelihood over μ_{mj} gives

$$\hat{\mu}_{mj} = \frac{1}{n}\sum_i \Big(y_{mij} - \sum_k \bar{l}_{ik}\bar{f}_{mjk}\Big), \tag{21}$$

and every block is re-centered, $\tilde{Y}_m = Y_m - \mathbf{1}\hat{\mu}_m^\top$. The survival term does not depend on μ (B.1). The intercept is not updated within the inner sweeps. During the removal check it is re-estimated for each candidate set of factors, because deleting a nonnegative-activity factor moves its mean contribution into the intercept.

B.4 Projection and predictor moments

Under the mean-field posterior the loadings of different features are independent, so

$$E(Z_{ik}) = \sum_{m,j} \tilde{y}_{mij}\bar{f}_{mjk}, \qquad \mathrm{Var}(Z_{ik}) = \sum_{m,j} \tilde{y}_{mij}^2 \,\mathrm{Var}(f_{mjk}), \tag{22}$$

and, with β independent of F,

$$E(\eta_i) = \sum_k E(Z_{ik})\bar{\beta}_k, \qquad \mathrm{Var}(\eta_i) = \sum_k \left[E(Z_{ik}^2)E(\beta_k^2) - E(Z_{ik})^2\bar{\beta}_k^2\right]. \tag{23}$$

The implementation computes $\mathrm{Var}(\eta_i)$ in the equivalent form $\sum_k[\mathrm{Var}(Z_{ik})E(\beta_k^2)+E(Z_{ik})^2\mathrm{Var}(\beta_k)]$. The per-sweep diagnostic stored as `fixed_working_objective` uses a different expression that omits cross-factor terms of $E(\eta_i^2)$; it is recorded for monitoring only and does not enter convergence, removal or any result in this report.

B.5 Shared activities

η does not depend on L, so only the reconstruction terms of both blocks enter. With $R_m^{(-k)} = \tilde{Y}_m - \sum_{l\neq k}\bar{L}_{\cdot l}\bar{F}_{ml}^\top$,

$$A_{L,k} = \sum_m \sum_j \tau_{mj}\overline{f^2}_{mjk}, \qquad B_{L,ik} = \sum_m \sum_j \tau_{mj} R_{mij}^{(-k)}\bar{f}_{mjk}. \tag{24}$$

The posterior for each l_{ik} under the point-exponential prior, with $x = B/A$ and $s^2 = 1/A$, is a mixture of a point mass at zero and a normal distribution with location $x - \lambda s^2$ truncated to $[0,\infty)$, with mixture weight proportional to $\pi\lambda\exp(-\lambda x + \lambda^2 s^2/2)\Phi((x-\lambda s^2)/s)$ against $(1-\pi)N(x;0,s^2)$. After the column update, $(\pi_{L,k}, \lambda_{L,k})$ is re-estimated by maximizing the marginal likelihood of the n pseudo-observations (`ebnm_point_exponential`, mode 0). Unlike a moment update, this fit can return the point mass ($\pi = 0$), which removes the column.

B.6 Modality-specific loadings

Reconstruction terms. As in Equation 13,

$$A_{mjk}^{\mathrm{rec}} = \tau_{mj}\sum_i \overline{l^2}_{ik}, \qquad B_{mjk}^{\mathrm{rec}} = \tau_{mj}\sum_i \bar{l}_{ik}R_{mij}^{(-k)}.$$

Survival terms and feature coupling. Write $\eta_i = \beta_k Z_{ik} + \eta_{i,-k}$ and $Z_{ik} = \tilde{y}_{mij} f_{mjk} + Z_{ik}^{(-j)}$, where $Z_{ik}^{(-j)}$ collects every other feature of every modality. Substituting into the expected surrogate Equation 20 and keeping the terms in f_{mjk},

$$f_{mjk}\Big[\bar{\beta}_k \sum_i \tilde{y}_{mij}(h_i - w_i E\eta_{i,-k}) - E(\beta_k^2)\sum_i w_i \tilde{y}_{mij} E(Z_{ik}^{(-j)})\Big] - \tfrac{1}{2} f_{mjk}^2\, E(\beta_k^2)\sum_i w_i \tilde{y}_{mij}^2.$$

The coefficient of f_{mjk}^2 uses $E(\beta_k^2)$ because β_k multiplies Z_{ik} in a squared term. Adding the reconstruction terms with equal weight,

$$A_{mjk} = \tau_{mj} \sum_i \overline{l^2}_{ik} + E(\beta_k^2)\sum_i w_i \tilde{y}_{mij}^2, \tag{25}$$

$$B_{mjk} = \tau_{mj} \sum_i \bar{l}_{ik} R_{mij}^{(-k)} + \bar{\beta}_k \sum_i \tilde{y}_{mij} h_{i,-k} - E(\beta_k^2)\sum_i w_i \tilde{y}_{mij} E(Z_{ik}^{(-j)}), \tag{26}$$

with $h_{i,-k} = h_i - w_i E(\eta_{i,-k})$. The last term couples feature j to all other features through the current projection: the survival information a feature receives depends on how much of the factor's projection the other features already explain. Features are therefore updated sequentially (Gauss–Seidel). After each feature, $E(Z_{\cdot k})$ and $\mathrm{Var}(Z_{\cdot k})$ are updated by the change in that feature's moments, $E(Z_{ik}) \mathrel{+}= \tilde{y}_{mij}(\bar{f}_{mjk}^{\text{new}} - \bar{f}_{mjk}^{\text{old}})$ and likewise for the variance. The survival terms are gated by $\bar{\beta}_k$ and $E(\beta_k^2)$: a factor with a coefficient near zero receives almost no survival information in its loadings (S38).

Point-Laplace posterior. With $x = B/A$, $s^2 = 1/A$, and prior $(1-\pi)\delta_0 + \pi\frac{\lambda}{2}e^{-\lambda|f|}$, the posterior is a mixture of a point mass at zero, a positive part distributed as $N(x - \lambda s^2, s^2)$ truncated to $f > 0$, and a negative part distributed as $N(x + \lambda s^2, s^2)$ truncated to $f < 0$. Their log weights are

$$\log(1-\pi) + \log N(x; 0, s^2), \qquad \log\tfrac{\pi\lambda}{2} + \tfrac{\lambda^2 s^2}{2} \mp \lambda x + \log\Phi\Big(\frac{\pm x - \lambda s^2}{s}\Big), \tag{27}$$

and the posterior mean and second moment follow from truncated-normal moments (with an asymptotic expansion when the truncation point is far in the tail). These closed forms agree with `ebnm` at a fixed prior to 10^{-6} (S01, 2026-10-01 entry). The point-Laplace parameters (π_{mk}, λ_{mk}) are held fixed during a modality–factor sweep and then re-estimated from that sweep's p_m pseudo-observations by `ebnm_point_laplace` with mode 0. The stored KL values are those of the prior used in the sweep.

B.7 Survival coefficients

After the loadings of factor k are updated in both modalities, the terms of the expected surrogate in β_k are $\beta_k \sum_i E(Z_{ik}) h_{i,-k} - \frac{1}{2}\beta_k^2 \sum_i w_i E(Z_{ik}^2)$, so

$$A_{\beta,k} = \sum_i w_i [E(Z_{ik})^2 + \mathrm{Var}(Z_{ik})], \qquad B_{\beta,k} = \sum_i E(Z_{ik})\, h_{i,-k}. \tag{28}$$

Unlike the single-modality update Equation 11, $A_{\beta,k}$ includes the projection variance, so loading uncertainty shrinks the coefficient, and there is no α weight. The prior variance is the single-observation marginal-likelihood estimate $\hat{\sigma}_k^2 = \max(0, x^2 - 1/A)$ with $x = B/A$. If $\hat{\sigma}_k^2 = 0$, then

$q(\beta_k) = \delta_0$; otherwise

$$\mathrm{Var}(\beta_k) = \left(A_{\beta,k} + \hat{\sigma}_k^{-2}\right)^{-1}, \qquad \bar{\beta}_k = \mathrm{Var}(\beta_k)\, B_{\beta,k},$$

and $\mathrm{KL} = \frac{1}{2}\left[(\mathrm{Var}(\beta_k) + \bar{\beta}_k^2)/\hat{\sigma}_k^2 - 1 - \log(\mathrm{Var}(\beta_k)/\hat{\sigma}_k^2)\right]$. The survival-active classification uses $|\bar{\beta}_k|/\mathrm{sd}(\beta_k) \geq 1.96$. The coefficients are initialized from a Cox fit on standardized initial projections, transformed back to the raw projection scale.

B.8 Precisions and expected squared residuals

After the inner sweeps,

$$\tau_{mj} = \frac{n}{\sum_i E_q(\tilde{y}_{mij} - l_i^\top f_{mj})^2}, \qquad E_q(\tilde{y}_{mij} - l_i^\top f_{mj})^2 = (\tilde{y}_{mij} - \bar{l}_i^\top \bar{f}_{mj})^2 + \sum_k (\overline{l^2}_{ik}\overline{f^2}_{mjk} - \bar{l}_{ik}^2 \bar{f}_{mjk}^2). \tag{29}$$

These are unregularized maximum-likelihood values. Without an intercept, a spare factor could fit a single feature almost exactly and send its τ_{mj} toward infinity, which made such factors impossible to remove; with the intercept this did not recur at moderate signal-to-noise (S01, 2026-10-01 entry). One precision per modality and a Gamma prior on τ_{mj} are implemented alternatives that were not used in the evaluated fits.

B.9 Scale canonicalization

For any $c_k > 0$, the map $(l_{\cdot k}, f_{mk}, \beta_k) \mapsto (c_k l_{\cdot k}, f_{mk}/c_k, c_k\beta_k)$ leaves $LF_m^\top$ and η unchanged. The likelihood therefore does not fix the scale of a factor, and coordinate updates can move along this direction; in the single-modality model with survival in the loading update this caused a runaway (S38). After each factor's coefficient update the implementation sets $c_k = \mathrm{sd}(EZ_{\cdot k})$ over training patients and applies the map, transforming the other quantities consistently:

$$\overline{l^2}_{\cdot k} \mapsto c_k^2\, \overline{l^2}_{\cdot k}, \quad \overline{f^2}_{mk} \mapsto \overline{f^2}_{mk}/c_k^2, \quad E(\beta_k^2) \mapsto c_k^2 E(\beta_k^2), \quad \lambda_{L,k} \mapsto \lambda_{L,k}/c_k, \quad \lambda_{mk} \mapsto c_k\lambda_{mk}. \tag{30}$$

(A normal loading prior variance would map to σ^2/c_k^2.) The prior rates transform so that the prior probability of each transformed coordinate is unchanged. After canonicalization, $\mathrm{sd}(EZ_{\cdot k}) = 1$ in training, so a coefficient is per training standard deviation of the joint projection. The step does not change the fitted reconstruction or predictor, but it is applied between coordinate updates and therefore changes the path of the algorithm.

B.10 Stopping rule

At the end of each outer sweep the algorithm computes, for each modality, $\max|\Delta(\bar{L}\bar{F}_m^\top)| / \max(1, \max|\bar{L}\bar{F}_m^\top|)$ and the same quantity for $E(\eta)$, and stops when all are below 10^{-4}. These quantities are invariant to the scale exchange of B.9, which is why they are used instead of changes in posterior moments. The cap is 500 outer sweeps. In the real-data sweep only the fits at starting ranks 3 and 4 met the criterion (S28).

B.11 Removal check and refitting

When the coordinate loop ends, whether by meeting the criterion or by reaching the cap, the removal check computes, for each factor k,

$$\Delta_k = \mathcal{E}(\text{factors} \neq k) - \mathcal{E}(\text{all factors}), \tag{31}$$

where $\mathcal{E}$ is Equation 6 evaluated with factor k's activities, loadings and coefficient set to zero, its KL terms removed, the other factors' posteriors held fixed, and μ and τ re-estimated. If $\max_k \Delta_k \geq 0$, the factor with the largest Δ_k is deleted, the algorithm restarts from the remaining factors' moments and priors (a warm start with $K-1$ factors), and the check repeats after that fit. Holding the other factors fixed makes the check conservative: a deletion is accepted only if the evidence does not fall even before the remaining factors adapt. The removal history records whether each fit was converged when it was checked.

The survival component of $\mathcal{E}$ is $\log \mathrm{PL}(E\eta)$, whereas the exact variational objective would use $E_q \log \mathrm{PL}(\eta)$. Because the log partial likelihood is concave in η, Jensen's inequality gives $E_q \log \mathrm{PL}(\eta) \leq \log \mathrm{PL}(E\eta)$: the plug-in overstates the survival term, by an amount that grows with $\mathrm{Var}(\eta)$. A second-order correction gives $\log \mathrm{PL}(E\eta) - \frac{1}{2}\sum_i w_i \mathrm{Var}(\eta_i)$, the `corrected` option; in the simulation sweep it changed no summary by more than 0.04 (Section 3.4). Neither form makes $\mathcal{E}$ a guaranteed lower bound on the marginal likelihood, and the coordinate updates are not guaranteed to increase it. On real data the reconstruction component, which sums over every feature of both modalities, dominates $\mathcal{E}$, so survival has little influence on which factors are kept.

B.12 Prediction

For a new patient with features aligned to the training features, $\eta_{\text{new}} = \sum_m y_{m,\text{new}}^\top \bar{F}_m \bar{\beta}$, using raw (uncentered) data; centering at the training intercept would shift every score by the same constant. Orientation is fixed from training (`flip = FALSE`): higher η means higher risk.

B.13 Base derivation versus evaluated framework

Table 13: Base multimodal derivation and the evaluated framework. Sources: S08, S29, S33, S34.

Component	September base derivation (S33)	Evaluated framework	Where
Intercept	None; nonnegative data scale required	μ_{mj} re-estimated each outer sweep and for each removal candidate	B.3
Loading prior	Point-exponential ($f \geq 0$)	Point-Laplace (signed) per modality and factor	B.6
Prior estimation	Moment (EM-type) updates	Marginal likelihood via `ebnm`; can return the point mass	B.5, B.6
Loading update	Equal-weight reconstruction and survival, sequential, with feature coupling	Same	B.6
Coefficient update	Uses $E(Z^2)$; normal prior with fitted variance	Same	B.7
Scale	Not fixed during fitting; columns may be rescaled for reporting	Canonicalized to $\mathrm{sd}(Z_{\cdot k}) = 1$ after each factor update	B.9
Stopping rule	Stabilization of the predictor and posterior moments	Relative change in reconstruction and $\eta < 10^{-4}$; cap 500	B.10

Component	September base derivation (S33)	Evaluated framework	Where
Rank	Fixed K	Removal check with plug-in survival term, then warm-start refit	B.11
Survival term in objective	Working surrogate	Plug-in $\log \mathrm{PL}(E\eta)$ for removal; surrogate for updates	B.2, B.11

Appendix C. Supporting exhibits

These exhibits are referenced from the main text and placed here to keep the main text within its page target.

Table 14: Biological characterization methods. "Unadjusted" means the saved p-values carry no correction for the number of tests.

Analysis	Input object	Population	Statistic and settings	Multiplicity	What the output supports
Ranked enrichment (fgsea)	Nonnegative posterior means $E(F)$ over the 2,064 selected genes	Established single-modality fit; one program at a time	Genes ranked by raw loading; `scoreType = "pos"` (one-sided, high loadings); set sizes 10–500; seed 1. Collections: Hallmark, Reactome, KEGG, GO-BP, and a custom PDAC collection (Moffitt and Bailey subtype lists and DeSurv's three factor lists). Normalized enrichment score (NES) compares a set's running-sum statistic with permutations; leading-edge genes are the set members ranked before the running-sum maximum.	Benjamini–Hochberg within each program × collection; reported at padj < 0.10. No control across programs or collections.	Association of high-loading genes with predefined sets; not mechanism or survival-driven discovery. S21, S22, S41, S42
Over-representation sensitivity	Top 50, 100, or 150 genes by $E(F)$	Same selected gene universe	Hypergeometric test (`enricher`); background is the 2,064 selected genes, not the genome. Checks dependence on the cutoff.	Adjusted p-values within each call; no global error control.	Threshold-based cross-check on the same universe; not independent replication. S21, S41, S42
Subtype concordance	Training latent activities $E(L)$, not projection scores	TCGA, $n = 144$	Spearman correlation with PurIST basal-like probability; Kruskal–Wallis test across PurIST calls	Unadjusted	Within-training correspondence to an established subtype axis. S23, S41, S53
DeSurv gene-list overlap	Top 270 genes by $E(F)$; DeSurv factor lists	Both lists restricted to the selected shared-gene background	Jaccard similarity; hypergeometric overlap test	Unadjusted	Gene-list correspondence; not a predictive comparison. The custom fgsea collection reuses the same lists. S35, S41

Analysis	Input object	Population	Statistic and settings	Multiplicity	What the output supports
External signature association	Mean of within-cohort z-scored expression over the available leading-edge genes of one selected set	Five held-out cohorts	Univariable Cox HR per unit of the mean signature score. C2 scores each program's lowest-padj set; C4 excludes the whole PDAC_custom collection for the two genomics-only programs. Gene coverage reported.	Unadjusted; C4 is a post hoc correction	Descriptive external association of the selected signature, not necessarily of the whole program; not preregistered confirmation. S24, S40, S41, S51, S52
Program directions and survival curves	Marginal association of each program's projection for direction labels; frozen Program 3 and 7 projections and overall risk for curves	Direction labels from the dated decision record; five external cohorts for curves	Within-cohort tertiles; pooled program curves with cohort-stratified log-rank tests. Historical D4 risk $= -\eta$.	Descriptive	Direction and held-out separation, without clinical cutoffs. Distinct from signature scores and from $E(L)$. S01, S36, S49
Multimodal top-gene interpretation	Signed expression loadings: top 8 positive and top 6 negative genes	Matched TCGA fit, starting rank 7	Manual reading of the gene lists; no fgsea or over-representation analysis	No test	Provisional expression-supported labels; weaker than the single-modality characterization. S31, S43
Multimodal modality shares	Posterior-mean L and modality-specific F_m	Same fitted programs	$\|\bar{L}_{\cdot k}\|^2 \|\bar{F}_{mk}\|^2$ divided by the sum of these rank-one squared norms within modality m	Descriptive	Relative fitted rank-one signal; not additive variance explained and not a survival contribution. S31, S43
Multimodal correspondence	Expression loading columns on shared genes	Joint fit versus expression-only fit, established D4 programs (1,781 shared genes), or an unsupervised `flash` fit of TCGA expression (14 factors, 3,000 genes)	Saved absolute Pearson correlations and best matches. Saved tables contain only $\|r\|$.	Descriptive	Pattern similarity with direction unavailable; independent best matches need not form a one-to-one alignment. S32, S43, S44

Multimodal marginal survival association	$Z_{\cdot k} = \sum_m \tilde{Y}_m \bar{F}_{mk}$, standardized within cohort	TCGA training and ICGC primary PDAC	Univariable Cox HR per one SD of the joint projection, which includes methylation	7 programs × 2 cohorts = 14 unadjusted tests	Descriptive marginal association, distinct from the joint survival-active classification. S31, S43

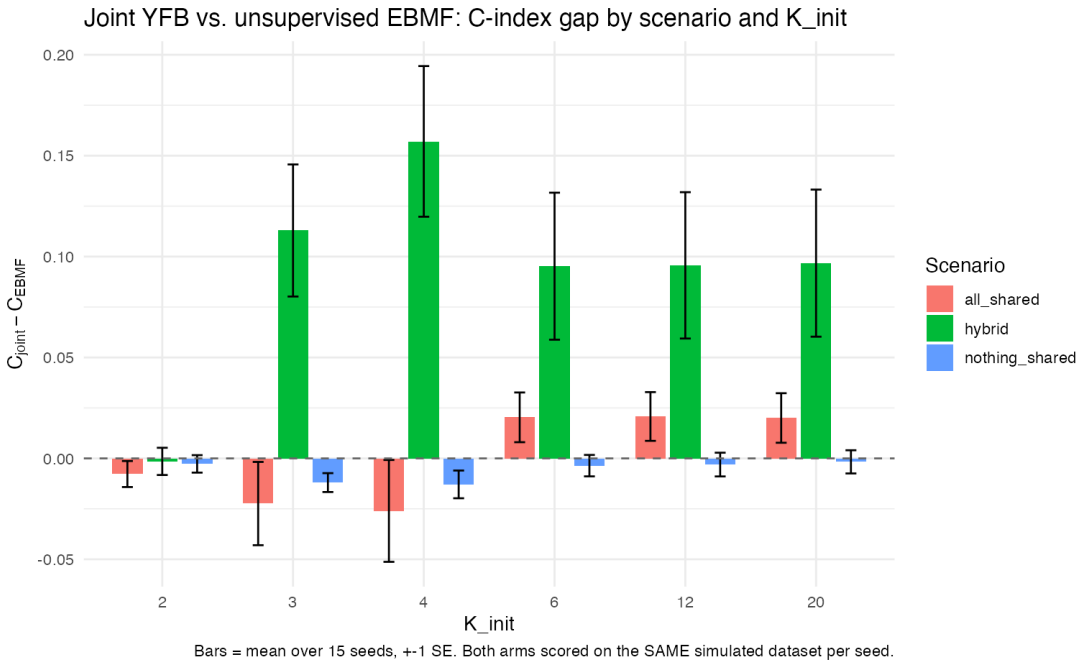


Figure 5: Difference in test concordance between the single-modality model and the two-step arm, by scenario and starting rank (true rank 6). Bars are means over 15 seeds with ± 1 standard error; both arms are scored on the same simulated data. Concordance is the outcome-oriented $\max(C, 1-C)$, and the two-step arm uses nonnegative `flashier` and unpenalized Cox. With $\alpha_F = 0$ the difference reflects the fitting procedure, not survival-guided programs. Source: S18; figure reused from the September 4 progress chapter.

Table 15: Single-modality simulation, means over 15 seeds at three of the six starting ranks (true rank 6). "Recovered" is the fraction of true programs matched at $|r| \geq 0.7$. "Active" is the fraction of true prognostic (or non-prognostic) programs whose matched estimate has $|\tilde{\beta}| > 0.001$. Test C is the outcome-oriented $\max(C, 1-C)$ on 75 test patients per seed. The two-step arm is nonnegative `flashier` followed by unpenalized Cox, not the principal real-data comparator. Source: S18.

Scenario	Starting rank	Shared recovered (model / two-step)	Specific recovered (model / two-step)	Prognostic programs active (model)	Non-prognostic active (model)	Test C (model / two-step)
All shared	4	0.59 / 0.54	–	0.60	–	0.744 / 0.770
All shared	6	1.00 / 0.89	–	1.00	–	0.873 / 0.853
All shared	20	1.00 / 0.89	–	0.93	–	0.872 / 0.852
Hybrid	4	0.80 / 0.27	0.57 / 0.65	0.80	0.23	0.772 / 0.615
Hybrid	6	1.00 / 0.73	0.93 / 0.83	1.00	0.35	0.825 / 0.730
Hybrid	20	0.97 / 0.73	0.98 / 0.83	1.00	0.27	0.826 / 0.729

Scenario	Starting rank	Shared recovered (model / two-step)	Specific recovered (model / two-step)	Prognostic programs active (model)	Non-prognostic active (model)	Test C (model / two-step)
Nothing shared	4	–	0.53 / 0.43	–	0.09	0.542 / 0.555
Nothing shared	6	–	0.96 / 0.70	–	0.10	0.541 / 0.545
Nothing shared	20	–	1.00 / 0.70	–	0.10	0.542 / 0.544